

1 Co-firing copper in a cordierite-type glass-ceramic
2 printed by multi-material vat photopolymerisation: a
3 model-based process design

4 [First Author]^{a,*}, [Second Author]^a

^a[Department, Institution], [City], [Country]

5 **Abstract**

Multi-material vat photopolymerisation can print a glass-ceramic body and its copper conductors as one green part, but whether such a part can be fired is not known. Carbon must leave the glass before its pores close, copper must stay metallic, and the two materials must shrink compatibly, all in one temperature range. A verified model couples pyrolysis and steam gasification of char, viscous sintering and crystallisation of the glass, diffusional sintering of copper, and co-firing mechanics, analytically and in three dimensions. Inputs come from the literature, and the glass was calibrated against the published IBM glass-ceramic/copper process. Of the two atmospheres compared, only steam with hydrogen held at a fixed fraction of the Cu/Cu₂O equilibrium, as in the IBM process, removes the char before the glass seals. Fine copper then densifies during the burnout, well before the glass, and the free-strain mismatch reaches 15.6%. A paste and programme optimised together use an assumed retarding effect of char on copper to release both materials at once, cutting the mismatch to 3.7%, but this design is fragile. A design that tolerates a factor of two in gasification rate and 15 K in the glass transition holds the mismatch at 9–11% and the copper at 89% IACS in a 16 h cycle. In

three dimensions the two designs reduce the dishing of a printed test vehicle from 622 μm to 150–317 μm , and plastic flow of the copper keeps the cooling bow small. Gasification kinetics, glass viscosity, copper sintering viscosity and char yield should be measured first.

6 *Keywords:* Co-firing, Glass-ceramic, Copper metallisation, Vat
7 photopolymerisation, Binder removal, Sintering mismatch

8 **1. Introduction**

9 Packages for millimetre-wave front ends, antenna-in-package modules
10 and wide-bandgap power devices now have to route conductors in three
11 dimensions through a low-loss dielectric, and to do it close to the die [1,
12 2, 3, 4]. Co-fired ceramics answer part of this demand. Low-temperature
13 co-fired ceramic (LTCC) and glass-ceramic substrates combine low dielectric
14 loss, hermeticity and a thermal expansion close to that of silicon [5, 6].
15 They are, however, built from stacked tapes: conductors are printed in
16 planes, interconnections run vertically through punched vias, and every
17 change of geometry requires new tooling [7, 5]. The Heterogeneous Integration
18 Roadmap devotes a chapter to additive manufacturing for packaging, including
19 lithography-based manufacturing of ceramic parts [8].

20 The conductor of choice in fired packages has long been silver, because it
21 can be co-fired in air. Silver migrates under bias and humidity and diffuses
22 into the glass when fired in oxidising conditions [9, 10, 11]. Copper conducts
23 almost as well, costs less and is far less prone to migration, but it must be
24 fired in an atmosphere that keeps it metallic. In such an atmosphere the
25 organic binder no longer burns off cleanly and part of it is left as carbonaceous

26 residue [12]; carbon sealed in a glass that has closed its pores can bloat or even
27 burst the part [13]. IBM solved this for its glass-ceramic/copper multilayer
28 substrates with a steam-based co-sintering process that gave a dense glass-
29 ceramic with high-conductivity copper [14, 6]. That solution was developed
30 for laminated tapes.

31 Vat photopolymerisation of ceramic slurries, in particular lithography-
32 based ceramic manufacturing (LCM), produces dense ceramics with features of
33 a few tens of micrometres [15, 16, 17]. It has already been used for monolithic
34 microwave filters and antenna substrates [18, 19, 20], and photocurable LTCC
35 suspensions have been formulated for it [21]. In these parts the metallisation is
36 still added afterwards, by plating or by printing on the fired ceramic [18, 22], or
37 the conductors are jetted into ceramic layers by a separate route [23, 24, 25, 26].
38 Multi-material vat photopolymerisation removes that restriction. Printers
39 such as the CeraFab Multi 2M30 switch between slurries within a layer
40 [27, 28, 29]. Co-sintered ceramic-ceramic parts have been demonstrated [30],
41 and zirconia and titania green bodies printed by vat photopolymerisation
42 have been assembled and co-sintered in a reducing atmosphere, which made
43 the titania conductive [31]. Copper slurries for vat photopolymerisation have
44 been developed separately [32, 33, 34, 35]. In principle, then, a glass-ceramic
45 body and its copper wiring can be printed as one green part. Copper and
46 LTCC suspensions have already been inkjet-printed into a three-dimensional
47 part and co-fired [23]. For vat photopolymerisation, whose slurries carry far
48 more binder, we found no study of how such a part should be fired, and no
49 model of co-firing that includes binder removal.

50 Four requirements make that firing difficult, and they are coupled. First,

51 the char from both slurries must be removed before the glass seals its pores,
 52 in an atmosphere that does not oxidise the copper. Second, the two materials
 53 must shrink at compatible times and by compatible amounts; otherwise
 54 the part cambers, delaminates or cracks around its conductors [36, 37, 38].
 55 Third, the glass must densify before it crystallises, because crystals stop
 56 viscous flow. Fourth, a thermal-expansion mismatch of about 14 ppm K^{-1}
 57 between copper and a cordierite-type glass-ceramic must be accommodated
 58 on cooling. In tape-based LTCC the second requirement is usually met by
 59 adjusting the metal paste, for example with ceramic additions [39], or the
 60 heating schedule, guided by master sintering curves [40]. Continuum models
 61 of co-sintering reproduce camber and mismatch stresses in ceramic and metal-
 62 ceramic bilayers [41, 42, 43, 44, 45, 46], but none includes the chemistry of
 63 binder removal, and none treats copper with a crystallising glass. The first
 64 and third requirements are therefore usually handled separately from the
 65 second, although in practice they compete for the same temperature range.

66 This paper sets out a model-based design of that firing for a copper
 67 conductor printed inside a cordierite-type glass-ceramic body. We extend a
 68 slab model of binder removal and sintering, which we developed for single-
 69 material printed copper (Appendix A), to a crystallising glass-ceramic. The
 70 two materials are coupled through the viscous analogue of laminate theory
 71 and through a three-dimensional finite-element model of a voxelised printed
 72 part. The model is verified against closed-form solutions and parameterised
 73 from published data, with every input tagged by its provenance. We use
 74 it for four purposes. We (i) map the atmosphere and temperature window
 75 in which carbon can be removed from the glass before its pores close while

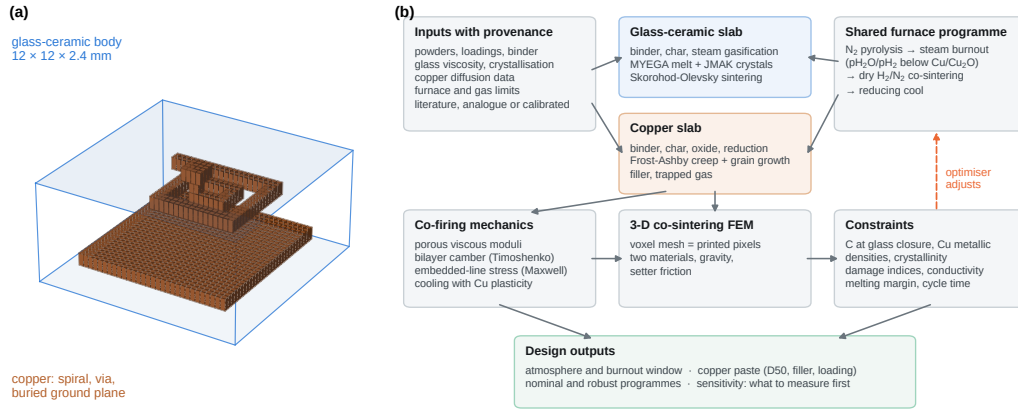


Figure 1: (a) Test vehicle: a $12 \times 12 \times 2.4$ mm glass-ceramic body with a buried copper ground plane, a square spiral and a via to a top pad, printed as one part. (b) Structure of the model. Each material is followed by its own slab model under a shared furnace programme; the two histories are coupled mechanically, analytically and in three dimensions, and the programme and paste are searched against constraints.

76 copper stays metallic; (ii) show that fine copper and a fragile glass densify in
 77 different temperature ranges and with different curve shapes, and quantify the
 78 consequences for camber and conductor damage; (iii) design the copper paste
 79 and the firing programme together to bring the two into step, both at nominal
 80 inputs and robustly against their uncertainty, and follow the designs through
 81 a printed test vehicle in three dimensions; and (iv) identify the few material
 82 parameters that decide the outcome, and should therefore be measured first.

83 2. Model formulation

84 2.1. Structure of the model

85 The co-fired part is represented at two levels (Fig. 1b). At the first level,
 86 each material is followed through the firing programme by a one-dimensional

87 slab model that resolves heat transfer, binder decomposition, char formation
88 and removal, pore closure and densification through the thickness of its critical
89 section. The glass-ceramic body and the copper conductor are simulated
90 separately but share the furnace set point and the inlet gas, so both see the
91 same thermal and chemical history. At the second level, the two densification
92 histories are coupled mechanically: analytically, through the viscous analogue
93 of laminate theory, and numerically, through a three-dimensional finite-element
94 model in which both materials occupy the voxels of the printed part. Both
95 levels use the same constitutive law for the porous body, so a free sample in
96 the finite-element model reproduces the slab model exactly.

97 The copper slab model was developed and verified for single-material
98 copper parts printed from a lithography-based slurry; its chemistry and its
99 calibration against published copper data are summarised in Appendix A. The
100 additions made here are the crystallising glass-ceramic, the copper paste with
101 an inert filler, the steam burnout atmosphere, and the co-firing mechanics.

102 2.2. Binder removal and residual carbon

103 The photocured binder is lumped into three pseudo-components i (a non-
104 reactive solvent, a crosslinked network and a char-forming backbone), each
105 decomposing by first-order kinetics,

$$\frac{dB_i}{dt} = -k_i(T) B_i, \quad k_i(T) = \frac{E_i \beta_{\text{ref}}}{RT_{\text{p},i}^2} \exp \left[-\frac{E_i}{R} \left(\frac{1}{T} - \frac{1}{T_{\text{p},i}} \right) \right], \quad (1)$$

106 where $T_{\text{p},i}$ is the temperature of the corresponding peak in a thermogravimetric
107 trace at $\beta_{\text{ref}} = 10 \text{ K min}^{-1}$. Writing the rate constant in terms of the peak
108 temperature keeps the two fitted quantities nearly independent and lets them
109 be read directly from a single thermogravimetric curve. A fraction χ of the

110 network and backbone mass remains as char. In a steam-bearing atmosphere
 111 the char is gasified by $\text{C} + \text{H}_2\text{O} \rightarrow \text{CO} + \text{H}_2$ at the rate

$$r_g = a(\varepsilon_o) (1 - f_{\text{cl}}) k_g(T) \frac{c}{M_{\text{C}}} \left(\frac{x_{\text{H}_2\text{O}}}{x_{\text{ref}}} \right)^{1/2} \times \frac{1 + K_{\text{H}} x_{\text{H}_2}^{\text{ref}}}{1 + K_{\text{H}} x_{\text{H}_2}} \left(1 - \frac{Q}{K_{\text{st}}(T)} \right)_+, \quad (2)$$

112 in which c is the char concentration, $a(\varepsilon_o) = \varepsilon_o/(\varepsilon_o + 0.02)$ the fraction of
 113 pore surface reachable through open porosity ε_o , f_{cl} the closed-pore fraction,
 114 K_{H} an inhibition constant for hydrogen adsorption, $Q = x_{\text{CO}}x_{\text{H}_2}/x_{\text{H}_2\text{O}}$ the
 115 reaction quotient and K_{st} the equilibrium constant from NIST-JANAF data
 116 [47]. Char in closed pores is not gasified: once the pores seal, whatever carbon
 117 remains stays in the part. The reference atmosphere is 4% H_2 in nitrogen
 118 humidified at a dew point of 20 °C, so that $x_{\text{ref}} = 0.023$ and $x_{\text{H}_2}^{\text{ref}} = 0.04$.
 119 The rate constant k_g is parameterised by a half-life $t_{1/2}$ at 800 °C in that
 120 atmosphere and an activation energy; no measured value exists for char on
 121 either material, so both enter the sensitivity analysis. The square of the
 122 Thiele modulus, $k_g L^2 / D_{\text{eff}} \sim 10^{-4}$ for $L = 2$ mm, shows that gasification
 123 is kinetically controlled, so steam access is represented by the open-pore
 124 accessibility $a(\varepsilon_o)$ rather than by a diffusion equation.

125 Pyrolysis gas raises the pore pressure. The overpressure is computed
 126 from the compressible Darcy equation with Kozeny–Carman and Knudsen
 127 contributions to the permeability and a polymer-permeation term that carries
 128 the flux before the pore network percolates; it is compared with the green or
 129 brown strength through a safety factor. The same slab tracks the through-
 130 thickness temperature difference and converts it into a thermal-stress index.

131 2.3. Densification

132 Both materials densify according to the Skorohod–Olevsky viscous sinter-
 133 ing model [48]. For a porous body of porosity θ and matrix shear viscosity η ,
 134 the stress is

$$\boldsymbol{\sigma} = 2\eta [\varphi(\theta) \dot{\boldsymbol{\epsilon}}' + \psi(\theta) \text{tr}(\dot{\boldsymbol{\epsilon}}) \mathbf{I}] + P_L(\theta) \mathbf{I}, \quad (3)$$

135 with $\varphi = (1 - \theta)^2$, $\psi = \frac{2}{3}(1 - \theta)^3/\theta$ and sintering stress $P_L = 3\gamma(1 - \theta)^2/r$,
 136 where γ is the surface energy and r half the Sauter mean particle diameter.
 137 A free body densifies at $\text{tr}(\dot{\boldsymbol{\epsilon}}) = -(P_L - \Delta p)/(2\eta\psi)$, where Δp is the pressure
 138 of insoluble gas trapped in closed pores. The two materials differ only in η , γ
 139 and r .

140 *Glass-ceramic.* The residual melt is Newtonian. Its viscosity follows the
 141 MYEGA equation [49],

$$\begin{aligned} \log_{10} \eta_g &= \log_{10} \eta_\infty + (12 - \log_{10} \eta_\infty) \frac{T_g}{T} \\ &\times \exp \left[\left(\frac{m}{12 - \log_{10} \eta_\infty} - 1 \right) \left(\frac{T_g}{T} - 1 \right) \right], \end{aligned} \quad (4)$$

142 with $\log_{10} \eta_\infty = -3$ (Pas), glass transition T_g (defined at 10^{12} Pas) and
 143 fragility index m . Crystallisation during firing follows non-isothermal Johnson–
 144 Mehl–Avrami–Kolmogorov kinetics [50] written in terms of the extended-
 145 volume integral,

$$\dot{\xi} = k_c(T) = k_{c,p} \exp \left[-\frac{E_c}{R} \left(\frac{1}{T} - \frac{1}{T_{p,c}} \right) \right], \quad X = 1 - \exp(-\xi^n), \quad (5)$$

146 which avoids the singularity of the rate form at $X = 0$ for $n > 1$. The constant
 147 $k_{c,p}$ is fixed numerically so that the simulated crystallisation exotherm at
 148 10 K min^{-1} peaks exactly at the measured temperature $T_{p,c}$; Kissinger’s closed

149 form [51] is exact only for $n = 1$ and would misplace the peak by about 1 K
 150 for $n = 2$. Crystals stiffen the melt as rigid inclusions,

$$\eta = \eta_g(T) \left(1 - \frac{X}{X_m}\right)^{-[\eta]X_m} \left(1 - \frac{f_C}{0.64}\right)^{-[\eta]0.64}, \quad (6)$$

151 the Krieger–Dougherty law [52] with intrinsic viscosity $[\eta] = 2.5$ and maximum
 152 packing X_m . The second factor treats residual char particles, volume fraction
 153 f_C , the same way. Once X approaches X_m the body stops flowing, which is
 154 how crystallisation arrests densification.

155 *Copper.* The copper skeleton creeps by diffusion. Its matrix viscosity combines
 156 lattice and grain-boundary diffusion [53],

$$\eta_{Cu} = f_\eta \frac{k_B T G^2}{42 \Omega (D_v + \pi \delta D_b / G)} \left(1 + \frac{C^2}{C_{inh}^2}\right) \left(1 - \frac{f}{0.64}\right)^{-1.6}, \quad (7)$$

157 where G is the grain size, Ω the atomic volume and f_η a calibration factor
 158 fitted to published copper densification data (Appendix A). The carbon term
 159 C (in ppm) describes a retarding effect of residual carbon on the sintering of
 160 copper. Its scale, $C_{inh} = 200$ ppm, is an estimate, and its influence on the
 161 results is examined in Section 4.10; and the last factor accounts for an inert
 162 ceramic filler of volume fraction f added to the conductor paste, a standard
 163 way to slow a metal paste down to the pace of its ceramic [39]. Grain growth
 164 is parabolic, driven by grain-boundary diffusion and pinned by porosity. The
 165 filler also lowers the conductivity; together with porosity,

$$\frac{\sigma_e}{\sigma_{e,0}} = (1 - \theta)^{3/2} \frac{1 - f}{1 + f/2}, \quad (8)$$

166 the second factor being the Maxwell Garnett result for insulating inclusions
 167 [54].

168 2.4. Atmosphere and the metallic state of copper

169 Steam removes char but oxidises copper once the water-to-hydrogen ratio
 170 exceeds the equilibrium of $\text{Cu}_2\text{O} + \text{H}_2 \rightleftharpoons 2\text{Cu} + \text{H}_2\text{O}$,

$$\frac{p_{\text{H}_2\text{O}}}{p_{\text{H}_2}} < K_{\text{Cu}}(T) = \exp\left(-\frac{\Delta G^\circ(T)}{RT}\right). \quad (9)$$

171 During the burnout segment the inlet hydrogen is therefore tied to the steam
 172 content, $x_{\text{H}_2} = x_{\text{H}_2\text{O}}/(s K_{\text{Cu}}(T_{\text{B}}))$, with a safety factor $s = 0.2$. The copper
 173 stays metallic by construction, and the steam is as oxidising towards carbon
 174 as the metal allows. This is the principle of the steam co-sintering developed
 175 for glass-ceramic/copper substrates [14, 6]; here it is applied in a model
 176 that says how long, at what temperature and in how much steam. Carbon
 177 gasification itself is thermodynamically unconstrained over the whole window
 178 (Section 4.2); its limits are kinetic.

179 2.5. Mechanics of co-firing

180 *Porous viscosities.* Eq. (3) describes an isotropic, linear viscous solid with
 181 shear viscosity $G_v = \eta\varphi$ and bulk viscosity $K_v = 2\eta\psi$, hence uniaxial viscosity
 182 $E_v = 9K_vG_v/(3K_v+G_v)$, viscous Poisson ratio $\nu_v = (3K_v-2G_v)/[2(3K_v+G_v)]$
 183 and biaxial viscosity $B = E_v/(1 - \nu_v)$. The free linear sintering rate is
 184 $\dot{\varepsilon}_{\text{f}} = -P_{\text{L}}/(6\eta\psi)$. The viscosities and the free rate follow from the slab
 185 histories of T , θ , G or ξ , and carbon. Shrinkage is reported as engineering
 186 linear shrinkage when single materials are compared, and as logarithmic strain
 187 in the co-firing histories, where the free-strain mismatch is the difference of
 188 the logarithmic strains of the two materials.

189 *Camber.* By the elastic–viscous correspondence principle, a bilayer of two
 190 linear viscous layers bends at a curvature rate given by Timoshenko’s bimetal
 191 solution [55], with elastic moduli replaced by biaxial viscosities and thermal
 192 strain by the difference in free sintering rates [56, 38, 41]:

$$\dot{\kappa} = \frac{6(1+\lambda)^2(\dot{\epsilon}_{f,2} - \dot{\epsilon}_{f,1})}{h[3(1+\lambda)^2 + (1+\lambda\mu)(\lambda^2 + 1/(\lambda\mu))]}, \quad (10)$$

193 with $\lambda = h_1/h_2$, $\zeta = B_1/B_2$ and $h = h_1 + h_2$. Layer 1 is copper and layer 2
 194 the glass-ceramic; positive curvature is concave towards the copper. Eq. (10)
 195 is integrated over the firing with the layer thicknesses updated by their own
 196 shrinkage. The reference stack is a 0.1 mm copper plane on 1 mm of glass-
 197 ceramic. Curvatures above about 100 m^{-1} lie outside the small-deflection
 198 range of the solution and are used only to compare designs.

199 *Embedded conductors.* A conductor line buried in the body cannot curl; its
 200 misfit is taken up by stress. We represent line and surrounding body as two
 201 Maxwell elements in series, each with a porous elastic modulus and a biaxial
 202 viscosity:

$$\dot{\epsilon}_{f,GC} - \dot{\epsilon}_{f,Cu} = \dot{\sigma} \left(\frac{1}{E_{Cu}} + \frac{1}{E_{GC}} \right) + \sigma \left(\frac{1}{B_{Cu}} + \frac{1}{B_{GC}} \right). \quad (11)$$

203 A compliant line follows the body and carries B_{Cu} times the misfit rate; a
 204 stiff line forces the body to flow around it. Tension in the copper ($\sigma > 0$)
 205 opens voids and breaks the line; compression in the copper means the body is
 206 shrinking onto a stiff inclusion and is itself in hoop tension, the classic origin
 207 of radial cracks around conductors [36, 37]. Two damage indices compare
 208 these stresses with the strength of the partner that fails at that moment,

$$\Pi_{Cu} = \frac{\max(\sigma, 0)}{\sigma_{s,Cu}(\rho_{Cu})}, \quad \Pi_{GC} = \frac{\max(-\sigma, 0)}{\sigma_{s,GC}(\rho_{GC})}, \quad (12)$$

209 where the strengths σ_s grow from the binder-carried green strength through
 210 the neck-controlled brown strength. The series coupling carries an Eshelby
 211 factor of order unity [57] that we do not resolve, so the indices are used
 212 comparatively. The line damage index reported below is the larger of the
 213 two, $\Pi_{\text{line}} = \max(\Pi_{\text{Cu}}, \Pi_{\text{GC}})$. The three-dimensional model below resolves the
 214 same coupling in full.

215 *Three-dimensional co-sintering.* The printed part is meshed voxel by voxel
 216 with trilinear hexahedra, so the mesh follows the layout of the printed part
 217 at the resolution of the voxel. Each element carries its material, porosity
 218 and microstructural state. The velocity field satisfies $\nabla \cdot \boldsymbol{\sigma} + \rho_b \mathbf{g} = \mathbf{0}$
 219 with Eq. (3), gravity, and a setter plate beneath the part: zero normal
 220 velocity and regularised Coulomb friction ($\mu_f = 0.6$) iterated to convergence
 221 at each step. Volumetric terms use one-point quadrature, which avoids
 222 locking as $\theta \rightarrow 0$. The problem is assembled with scikit-fem [58] and solved
 223 by conjugate gradients preconditioned with smoothed-aggregation algebraic
 224 multigrid carrying the six rigid-body modes [59]. Porosity is advanced with
 225 the exact mass-conserving update $1 - \theta^{n+1} = (1 - \theta^n) \exp(-\text{tr } \dot{\boldsymbol{\epsilon}} \Delta t)$. The
 226 temperature and carbon histories are taken from the two slab simulations.
 227 Below 0.3% porosity the constitutive law is evaluated at 0.3%, since the bulk
 228 viscosity otherwise grows without bound and the remaining densification no
 229 longer affects the shape. The simulation ends when the glass-ceramic reaches
 230 98% density or 90% crystallinity, whichever comes first; the remaining linear
 231 shrinkage of the body is then a few tenths of a per cent.

232 *Cooling.* Below a stress-free temperature T_{sf} , where the residual glass stops
 233 relaxing, the thermal-expansion mismatch is carried elastically by the glass-

ceramic and elastoplastically by the copper. The copper layer is treated as elastic-perfectly plastic with a temperature-dependent yield stress; its mean stress is capped at yield by a uniform plastic strain, and the curvature follows from Timoshenko’s elastic solution for the remaining elastic misfit.

2.6. Numerical solution, design search and sensitivity

The slab equations are integrated with a linearly implicit Rosenbrock method of orders 2(3) [60] on a geometrically graded mesh that is finest at the surface. A co-firing evaluation, two slab simulations and the mechanical post-processing, takes about 6 s. Designs were searched by differential evolution [61] over eleven variables: copper median particle size, filler fraction and solids loading; burnout temperature, time and steam fraction; temperature and duration of an intermediate matching hold; heating rate to the peak; peak temperature and hold. The objective was the largest free-strain mismatch between the two materials during firing, $\max_t |\varepsilon_{f,GC} - \varepsilon_{f,Cu}|$, which governs both camber and conductor stress. It was minimised subject to hard constraints and to a cycle-time budget of 18 h, the ε -constraint form of the trade-off between cycle time and mismatch. The constraints were: carbon at glass pore closure below 100 ppm and in the copper below 200 ppm; relative densities of at least 0.97 for the glass-ceramic and 0.92 for the copper; a crystallised fraction of at least 0.8; a line damage index below 1; no trapped-gas pressure above 1.2 times the sintering stress; and a positive margin to the melting point of copper. Violations entered the objective as a penalty proportional to their size, so the search is steered back towards feasible designs. Sobol initialisation rounds the population up to a power of two, so 128 designs were evolved for 15 generations after initialisation (2048 evaluations). The

259 objective still fell by 7% in the last generation, so the result is the best design
260 found rather than a proven optimum; it is called the nominal optimum below.

261 A second, robust search asked for a design that works across the uncertainty
262 of the two inputs that decide the timing, the gasification rate and the glass
263 transition. It evaluates each design at two corners of that uncertainty. At the
264 first, gasification is slow ($t_{1/2} = 4$ h) and the transition low ($T_g - 15$ K), so
265 the glass tends to seal before the char has gone. At the second, gasification is
266 fast ($t_{1/2} = 1$ h) and the transition high ($T_g + 15$ K), so the copper tends to
267 be released before the glass is ready. The objective is the larger of the two
268 peak mismatches; all constraints must hold at both corners, and the copper
269 must in addition reach 80% IACS. The cycle-time budget was relaxed to 20 h,
270 because the first feasible designs found needed about 20 h; the final design
271 needs 16 h, so the budget did not bind. The search was seeded with one such
272 feasible design, and a population of 128 was evolved for eight generations
273 (1152 evaluations, each over both corners). Only the two corners described
274 were used in the search; the other two corners were checked on the final
275 design. The nominal case lies between the corners and was checked on the
276 final design.

277 The influence of uncertain inputs was screened with Morris elementary
278 effects [62]: 16 trajectories through a four-level grid over fourteen factors,
279 240 co-firing evaluations in total, around the nominal optimum. Twelve
280 factors are the uncertain inputs with the ranges of Table 1; the other two are
281 manufacturing tolerances on the copper paste, -25 to $+33\%$ in particle size
282 and ± 0.05 in filler fraction. For each output the mean absolute elementary
283 effect μ^* measures the overall influence of a factor and the standard deviation

284 σ its non-linearity and interaction with other factors.

285 **3. Parameters and calibration**

286 *3.1. Provenance of the inputs*

287 No measurements were made for this study. Every input therefore comes
288 from one of five sources, and Table 1 states which one applies. A value is
289 labelled L when it was reported for the glass-ceramic of the IBM process or
290 for copper itself; A when it was taken from an analogous material, usually a
291 commercial copper-compatible LTCC; C when it was calibrated here against
292 published process observations (Section 3.2); D when it follows from other
293 entries of the table; and E when it is an engineering estimate. Most estimated
294 and analogue values that matter enter the sensitivity analysis of Section 4.10,
295 with the ranges given in the table; three that do not were checked one at a
296 time.

297 The reference glass is the MgO-rich cordierite-type glass used by IBM
298 for its copper-compatible substrates (glass #12: 52.5 SiO₂, 22.0 MgO, 22.0
299 Al₂O₃, 1.5 P₂O₅, 0.5 B₂O₃, 1.5 ZrO₂, wt%) [13, 63]. It was chosen because,
300 among copper-compatible glass-ceramics, it is the one for which the firing
301 process, the atmosphere and the reasons behind the chosen temperatures are
302 best documented. We could not obtain published viscosity data for it, so its
303 viscosity was calibrated.

304 *3.2. Calibration of the glass against the IBM process*

305 The patents that describe the IBM steam co-sintering process state, for
306 glass #12, why the burnout hold sits where it does. The hold is placed

307 at 785 ± 10 °C, between the annealing and softening points of the glass.
 308 At 750 °C carbon removal would take “prohibitively excessive amounts of
 309 time”; at 830 °C the glass pores close and trap water and binder residue
 310 [13]. A later IBM patent on the co-sintering of multilayer ceramic substrates
 311 states that 33–50% of the total fired shrinkage occurs during the burnout
 312 step [78]; we assume that this holds for glass #12, and a third places the
 313 onset of crystallisation at about 900 °C, after the burnout hold and before
 314 the crystallisation hold at about 960 °C [79]. We simulated the full worked
 315 schedule of Ref. [13] (N_2 to 200 °C, $\text{H}_2/\text{H}_2\text{O}$ to 450 °C, 2.9 K min^{-1} to the
 316 hold, 6 h at $\text{H}_2/\text{H}_2\text{O} = 10^{-4}$, 0.5 h in N_2 , 2.1 K min^{-1} to 960 °C, 2 h) and
 317 used these statements as targets. The patent specifies the $\text{H}_2/\text{H}_2\text{O}$ ratio but
 318 not the dilution; we assumed 50% steam in nitrogen.

319 Two of them fix the viscosity. Requiring that a 6 h hold at 780 °C
 320 produce 40% of the fired shrinkage with the pores still open, and that the
 321 pores close within 2 h at 830 °C, gives $\log_{10} \eta = 10.62$ at 780 °C and 9.37 at
 322 830 °C. The MYEGA curve through these two points, with $\log_{10} \eta_\infty = -3$,
 323 has $T_g = 733.6$ °C and $m = 32.4$. The remaining observations were not used
 324 in the fit and serve as checks. Between 800 and 860 °C the calibrated viscosity
 325 falls by one decade per 43 K. Giess et al. found that the sintering of an IBM
 326 cordierite-type glass over the same range followed the viscosity of the glass,
 327 which fell by about one decade per 40 K [65]. The annealing point, taken
 328 as 10^{12} Pa s, lies at 734 °C and the Littleton softening point ($10^{6.6}$ Pa s) at
 329 980 °C, so the 785 °C hold lies between them as the patent requires; so does
 330 the dilatometric softening point, near 10^{10} Pa s, which the calibrated glass
 331 reaches at about 803 °C. Stoichiometric cordierite glass is more viscous by 3.7

Table 2: The IBM worked schedule [13] simulated at three hold temperatures. Shrinkage is the fraction of the total fired shrinkage reached by the end of the 6 h hold; carbon is the residual carbon in the glass-ceramic at the end of the hold.

Hold temperature	750 °C	785 °C	830 °C
Shrinkage during hold	7%	47%	100%
Pores at end of hold	open	open	closed
Carbon at end of hold (ppm)	33	0.006	0.07
Final relative density	0.990	0.990	0.990
Final crystallised fraction	1.00	1.00	1.00
Statement in [13]	too slow	chosen	pores close

332 decades at 780 °C, 1.0 at 900 °C and 0.4 at 960 °C [80]. Glass #12 contains
 333 more MgO and less Al₂O₃ than cordierite, together with B₂O₃ and P₂O₅, so
 334 a lower viscosity is plausible, but this comparison is not a test of the fit.

335 The crystallisation kinetics were treated in the same way. With the acti-
 336 vation energy measured on an IBM-type glass ($E_c = 303 \text{ kJ mol}^{-1}$ [67]) and
 337 $n = 2$, placing the onset of crystallisation (5% crystallised) at 900 °C in the
 338 worked schedule requires an exotherm peak at 1040 °C at 10 K min^{-1} . The
 339 simulated exotherm then spans 978–1066 °C, a width of about 90 K, compa-
 340 rable to the 80–100 °C span of sintering temperatures that the IBM patent
 341 finds covered by the crystallisation exotherm of its glasses [63]. Differential
 342 thermal analysis of other MgO–Al₂O₃–SiO₂ glasses places the α -cordierite
 343 peak at 1006–1032 °C [64, 69].

344 Table 2 summarises the result. The 785 °C hold removes the char and
 345 leaves 47% of the shrinkage to the burnout step, inside the 33–50% reported
 346 for the process, while the pores stay open. At 830 °C the body densifies

347 completely during the hold and seals its pores. These two rows reproduce the
 348 calibration targets and are therefore not independent evidence. The 750 °C
 349 row is independent: it depends on the gasification kinetics, which were not
 350 fitted, and it leaves four orders of magnitude more carbon than the 785 °C
 351 hold. The model does not reproduce the absolute time scale of the patent’s
 352 statement. With the estimated gasification rate, 33 ppm of carbon remain
 353 after 6 h at 750 °C, and whether that would have been acceptable cannot be
 354 judged, because no carbon specification was published. Nor does the model
 355 reproduce the reason the patent gives for avoiding 830 °C: the carbon has
 356 gone before the pores close, and the trapped water that the patent mentions
 357 is not modelled. For all three holds the final body is 99% dense and fully
 358 crystallised, and its free linear shrinkage is 23.1%.

359 **4. Results**

360 *4.1. Verification*

361 The model was checked against closed-form solutions before it was used
 362 for design (Appendix B). Isothermal densification of the glass-ceramic follows
 363 the Skorohod–Olevsky solution to within 1.3×10^{-4} in porosity for 4 to 16
 364 cells (Fig. 2a), and the three-dimensional curvature rate of a free copper/glass-
 365 ceramic plate converges to the viscous Timoshenko value at least at second
 366 order, with an error of 0.05% at 0.15 mm voxels (Fig. 2b). The analytical
 367 camber of Eq. (10) and the finite-element model therefore describe the same
 368 mechanics, and the analytical form can be used inside the design search.

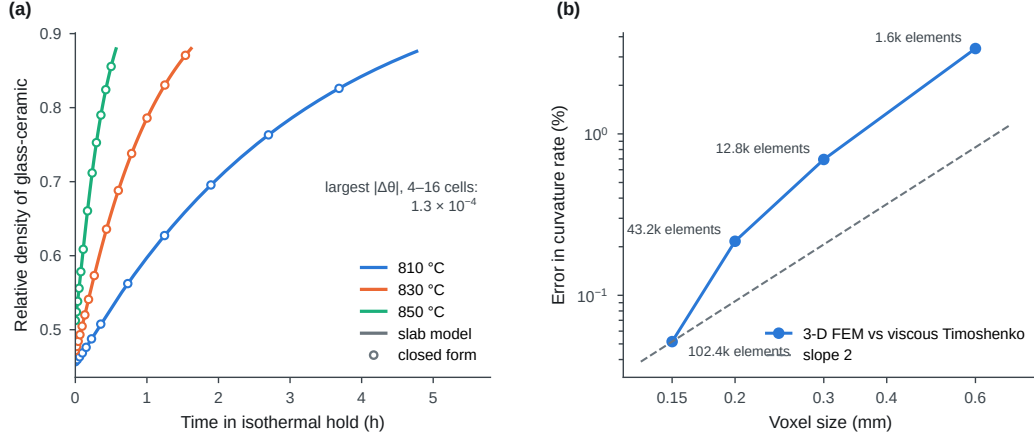


Figure 2: Verification. (a) Isothermal densification of the glass-ceramic slab at 810, 830 and 850 °C (lines, 8 cells) against the closed-form solution integrated along the simulated part temperature (circles); crystallisation and char were switched off. (b) Error of the three-dimensional curvature rate of a free 12×12 mm plate (0.6 mm copper on 1.8 mm glass-ceramic, 900 °C) against the viscous Timoshenko solution, Eq. (10), as the voxel size is refined.

4.2. Atmosphere window for burnout

Fig. 3 places the burnout atmosphere in the plane of temperature and $p_{\text{H}_2\text{O}}/p_{\text{H}_2}$. Copper remains metallic below the orange line, which falls from $10^{6.2}$ at 500 °C to $10^{4.8}$ at 780 °C and $10^{4.1}$ at 1000 °C. Carbon gasification by steam is thermodynamically possible above the dashed line, which lies eight or more decades lower. The two constraints therefore leave a wide band of permissible atmospheres, and the choice within it is kinetic. At 780 °C, 30% steam with hydrogen at 20% of the copper limit ($p_{\text{H}_2\text{O}}/p_{\text{H}_2} = 1.2 \times 10^4$) gives a char half-life of 0.43 h. Forming gas (4% H_2) humidified through a bubbler at 20 °C, a common choice for copper, contains 2.3% water; its half-life at the same temperature is 3.1 h, seven times longer, because the steam partial

380 pressure is low and hydrogen inhibits the reaction.

381 The operating point of the IBM process, $H_2/H_2O = 10^{-4}$ at 785 ± 10 °C
382 [13], lies at 17% of the copper limit and almost coincides with the steam
383 point in Fig. 3. The rule used here, hydrogen tied to steam so that the
384 water-to-hydrogen ratio sits at a fixed fraction of the copper limit, is therefore
385 consistent with the atmosphere used for tape-cast substrates. It also says
386 how the ratio should change if the burnout temperature changes: by a factor
387 of about 2.5 per 100 K, following $K_{Cu}(T)$.

388 *4.3. Removing carbon before the glass seals*

389 The char has to leave the glass before its pores close, and the glass
390 carries far more of it than the copper does. After pyrolysis in nitrogen the
391 glass-ceramic contains 22 500 ppm carbon relative to its solids, the copper
392 5 300 ppm. The difference follows from the densities: both slurries hold a
393 similar volume of binder, but a unit volume of glass-ceramic solids weighs
394 a third as much as the same volume of copper. Carbon on the glass is also
395 the more harmful, since whatever is sealed in reacts with the water released
396 during crystallisation and bloats the part [13].

397 Fig. 4 maps the carbon left in the glass-ceramic at the moment its pores
398 close, as a function of burnout temperature and hold time. In steam (Fig. 4a)
399 the 100 ppm contour runs from 16 h at 710 °C to 1.1 h at 830 °C, and the
400 window closes abruptly at 850 °C, where the glass seals during the hold. The
401 IBM hold of 6 h at 785 °C falls in the region with less than 10 ppm. In
402 humidified forming gas (Fig. 4b) only the longest hold, 16 h at 790 °C, brings
403 the carbon down to 350 ppm; every other combination leaves more than
404 1000 ppm. Residual carbon also slows the densification of the glass. Acting

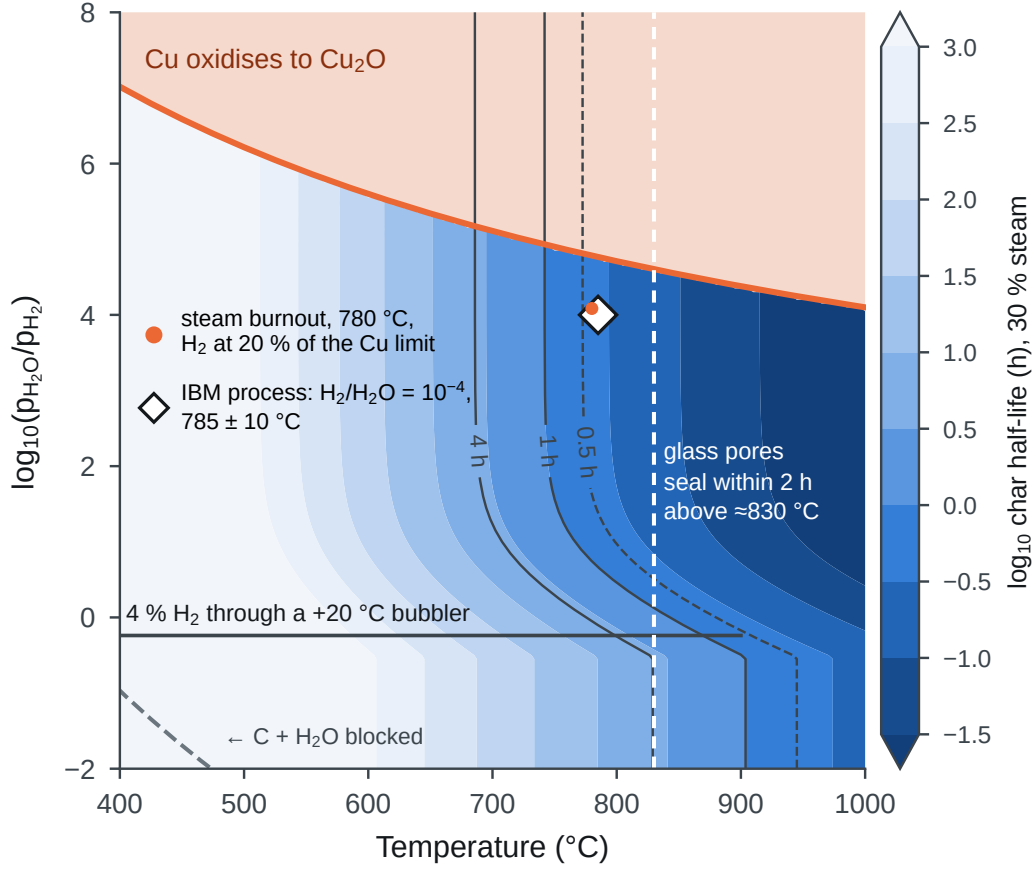


Figure 3: Burnout atmosphere window. Colour: half-life of char in 30% steam at the given $p_{\text{H}_2\text{O}}/p_{\text{H}_2}$ (reference kinetics of Table 1). Copper oxidises above the orange line; carbon cannot be gasified to the left of the dashed line ($p_{\text{CO}} = 10^{-4}$ atm). Circle: steam burnout used in this work. Diamond: operating point of the IBM process [13]. The white line marks the temperature above which a 2 h hold seals the pores of the calibrated glass.

405 as rigid inclusions, the 0.03–2% of carbon retained in forming gas lowers the
 406 final density from 0.99 to 0.96–0.99. For this material pair, steam is therefore
 407 necessary rather than merely helpful.

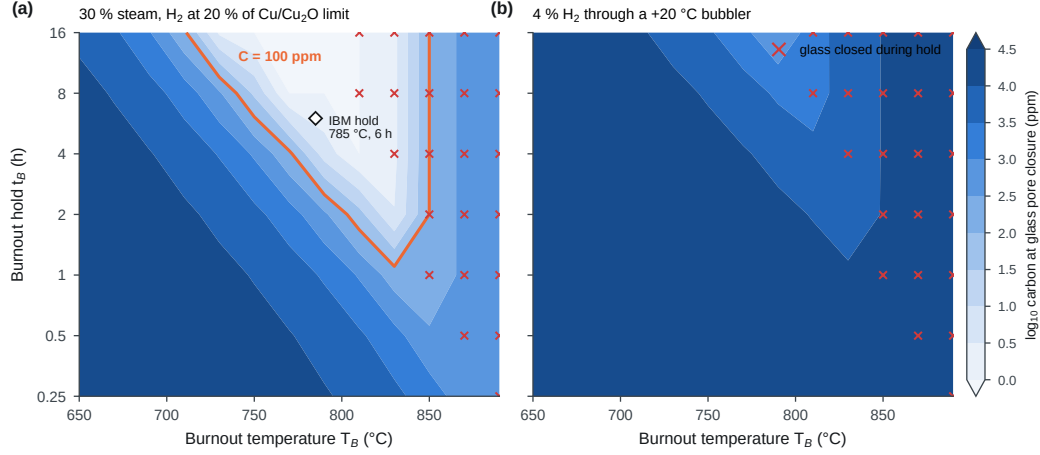


Figure 4: Carbon remaining in the glass-ceramic when its pores close, for burnout holds of temperature T_B and duration t_B within the baseline programme. (a) 30% steam with hydrogen at 20% of the Cu/Cu₂O limit; (b) 4% H₂ humidified at 20 °C. Orange: 100 ppm contour. Crosses: the glass sealed during the hold. Diamond: the IBM burnout hold [13].

4.4. Two densification ranges

Fig. 5 compares the free sintering of the two materials after binder removal at 5 K min⁻¹ in dry 4% H₂. The glass-ceramic shrinks by 23% between 850 and 933 °C (5 and 95% of its shrinkage) and then stops, because it crystallises: 93% of the glass has crystallised at 1060 °C. Fine copper densifies much earlier, and over a much wider range. A 3 µm powder shrinks between 454 and 702 °C; a 1 µm powder between 323 and 463 °C. Only for powders of 12 µm or coarser does the midpoint of densification fall inside the range of the glass, and these powders do not reach high density before 1060 °C: 83% for 12 µm and 60% for 20 µm.

The two curves also differ in shape. The glass densifies over a range of about 80 K, as a Newtonian melt whose viscosity falls steeply with tempera-

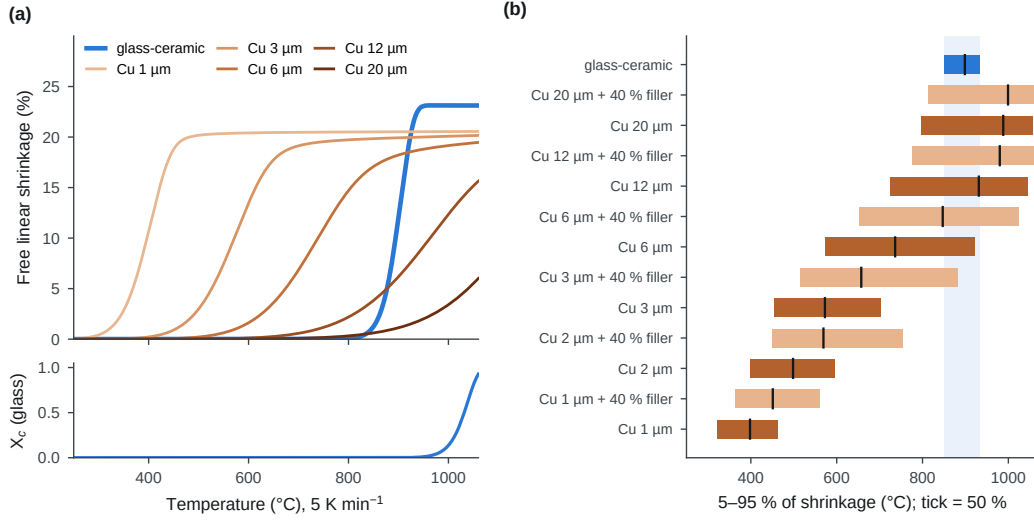


Figure 5: Free sintering of the two materials at 5 K min⁻¹ in dry 4% H₂ after binder removal. (a) Linear shrinkage of the glass-ceramic and of copper of different median particle size, and crystallised fraction of the glass. (b) Temperature range between 5 and 95% of the final shrinkage, with the midpoint marked, for copper with and without 40 vol% inert filler; the shaded band is the range of the glass-ceramic. Bars that reach 1060 °C had not finished densifying.

420 ture. Copper densifies over 140–350 K, because diffusional creep has a lower
 421 activation energy and grain growth slows it as it proceeds. Adding an inert
 422 filler to the paste delays and stretches the copper curve (Fig. 5b): 40% filler
 423 shifts the midpoint of 3 µm copper from 573 to 657 °C and widens the 5–95%
 424 range from 248 to 368 K. Neither particle size nor filler, on its own, gives
 425 copper a shrinkage curve that is as late and as steep as that of the glass. The
 426 matching has to come partly from the furnace programme.

4.5. *Co-firing with an unmatched paste*

The baseline design combines a 3 μm copper paste without filler with a programme that follows the IBM pattern: pyrolysis in nitrogen to 460 °C, a 4 h burnout at 780 °C in 30% steam, and sintering at 960 °C for 1 h in dry 4% H_2 (Fig. 6a). The design meets the chemical targets. The glass-ceramic retains 33 ppm of carbon when its pores close and reaches 99.1% density and full crystallinity; the copper is 97.6% dense, retains 81 ppm of carbon and conducts at 96% IACS.

It fails the mechanical ones. As soon as steam begins to remove the char, the copper starts to sinter, while the glass is still rigid. About an hour before the end of the burnout hold, the logarithmic shrinkage of the copper is 20.6% and that of the glass 4.9%. This is where the free-strain mismatch peaks, at 15.6%, and where a free bilayer of 0.1 mm copper on 1 mm glass-ceramic would reach a curvature of 58 m^{-1} , concave towards the copper. The damage index of an embedded line reaches $\Pi_{\text{Cu}} = 1.6$ at 680 °C, during the ramp to the burnout temperature, when the line is pulled in tension by the rigid body around it while its own necks are still weak. The glass then catches up, and its shrinkage overtakes that of the copper at 889 °C. The bilayer bends the other way and reaches -259 m^{-1} at 930 °C. The final logarithmic shrinkages differ by 4%: 26.3% for the glass-ceramic against 22.3% for the copper. For this material pair the difficulty lies less in the final shrinkage than in the order in which the two materials shrink.

4.6. *Copper paste design maps*

Fig. 7 shows how far the materials alone can repair the baseline, with the programme left unchanged. The upper row varies the copper paste. Coarser

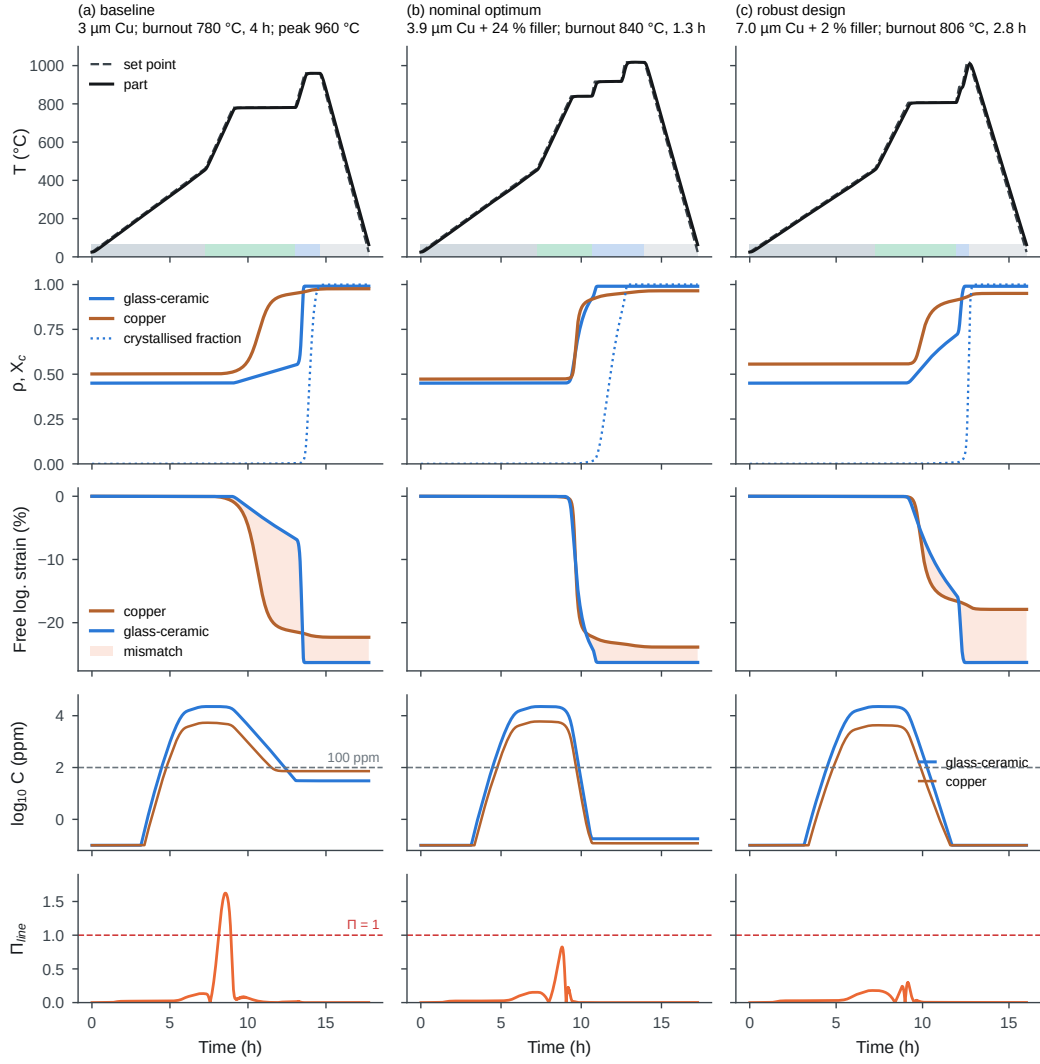


Figure 6: Co-firing histories of (a) the baseline, (b) the nominal optimum and (c) the robust design of Table 3. From top: furnace set point and part temperature, with the atmosphere of each segment shown by the band at the bottom (grey, N_2 ; green, steam; blue, dry H_2/N_2); relative densities and crystallised fraction of the glass; free logarithmic shrinkage of each material, the shaded area being the mismatch; carbon in the glass-ceramic and damage index of an embedded copper line.

powder and filler both delay the copper, and the damage index of an embedded
 line falls below 1 for particle sizes of 4.5 μm and above (Fig. 7b). The peak
 mismatch falls more slowly. Its lowest values, about 9.5%, are reached at
 6 μm with 30% filler and at 9 μm without filler; coarser pastes are released
 too late and the mismatch rises again, now with the copper behind the glass.
 Every step towards a later copper costs density or conductivity (Fig. 7c,d).
 Filler lowers the conductivity directly, by 14 points of IACS for every 10 vol%,
 while coarse powder lowers it through porosity: 18 μm copper reaches only
 76% density. Among the pastes that meet the density limit, 9 μm copper
 without filler gives the best compromise: 9.5% peak mismatch, a damage
 index of 0.38 and 91% IACS. None of them reduces the final camber, which
 stays between -190 and -260 m^{-1} for every paste that meets the density
 limit. The final camber is set by the difference in final shrinkage, which the
 paste barely changes.

The lower row varies the glass, with the baseline paste. The glass transition
 decides whether carbon is trapped. Below about 695 $^{\circ}\text{C}$ the glass seals during
 the 780 $^{\circ}\text{C}$ burnout and retains about 3000 ppm (Fig. 7e). Above that, the
 carbon stays below 100 ppm, but the higher the transition, the later the glass
 densifies, and the crystallisation peak has to rise with it if the glass is to reach
 full density before it crystallises (Fig. 7f). The crystallisation peak must not
 be too high either: above about 1080 $^{\circ}\text{C}$ the glass is less than 80% crystalline
 after 1 h at 960 $^{\circ}\text{C}$ (Fig. 7g). The most interesting region is the narrow band
 around $T_g = 700 \text{ }^{\circ}\text{C}$ (Fig. 7h). There the glass densifies during the burnout
 hold, together with the copper, and the peak mismatch falls to 6.5% while
 the carbon at closure stays near 50 ppm. A glass chosen to match the copper

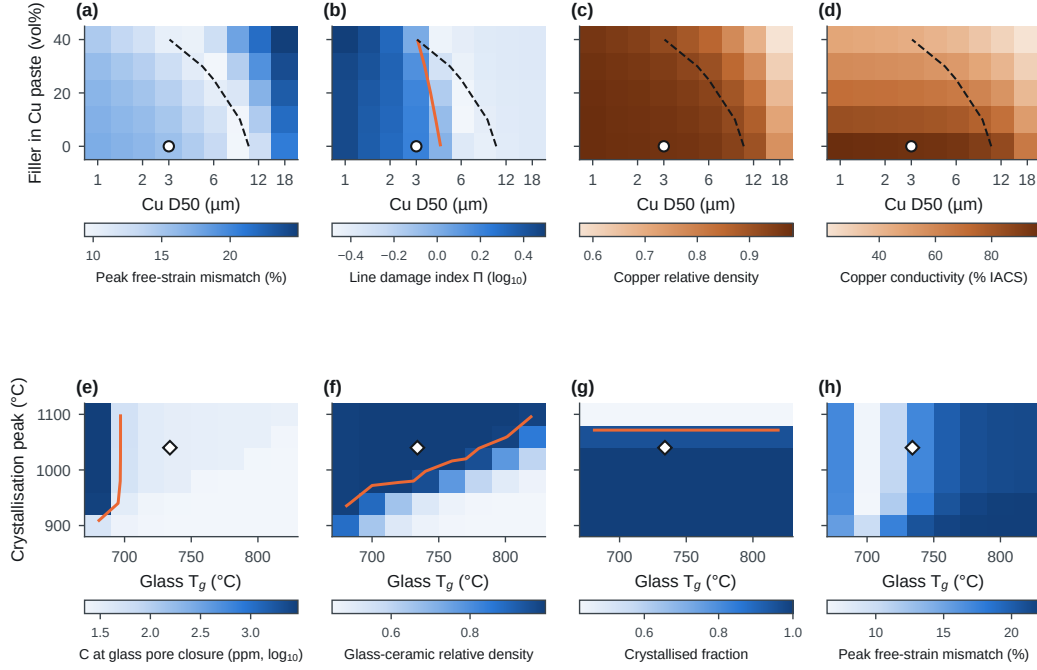


Figure 7: Design maps at the baseline programme. Top: copper paste, median particle size against filler fraction, with the baseline paste marked (circle): (a) peak free-strain mismatch; (b) damage index of an embedded line, orange contour $\Pi = 1$; (c) copper density; (d) conductivity. The dashed line is the density limit $\rho_{\text{Cu}} = 0.92$; pastes to its right fail it. Bottom: glass, transition temperature against crystallisation peak, with the calibrated glass marked (diamond) and the baseline paste: (e) carbon at pore closure, orange contour 100 ppm; (f) density of the glass-ceramic, orange contour 0.97; (g) crystallised fraction, orange contour 0.8; (h) peak mismatch.

477 would thus do more than any paste, but only within a window of about 20 K
 478 in T_g , and with little margin on carbon at its lower edge.

479 4.7. A matched paste and programme

480 Neither the paste nor the glass alone brings the peak mismatch below about
 481 6%, so the paste and the programme were searched together (Section 2.6).

482 The best design found within an 18 h cycle, called the nominal optimum
483 below, is listed in Table 3 and its history is shown in Fig. 6b. It uses a 3.9 μm
484 copper powder with 24 vol% filler. The burnout is short and hot: 1.3 h at
485 840 °C in 81% steam, just below the temperature at which the glass would
486 seal during the hold (Fig. 4a). A 1.7 h hold at 916 °C follows, and then 1.2 h
487 at 1018 °C. The peak mismatch falls from 15.6 to 3.7%, the damage index
488 of an embedded line from 1.6 to 0.83, and the final camber of the reference
489 bilayer from -254 to -76 m^{-1} . The glass-ceramic is 99.0% dense and fully
490 crystallised, with 0.2 ppm of carbon at pore closure.

491 The nominal optimum works through the retarding effect of carbon on
492 copper sintering that the model assumes (Eq. (7)). The copper cannot sinter
493 while it carries char (Section 4.5), so the burnout releases it. Placing the
494 burnout just below the coalescence of the glass, and making it fast with a
495 high steam fraction, releases the copper at the moment the glass begins to
496 flow. The two materials then densify together (Fig. 6b), and the filler slows
497 the copper just enough to follow the glass rather than lead it. The glass is
498 already 99% dense when the hold at 916 °C begins. The hold crystallises it,
499 while the copper continues to densify, and the peak at 1018 °C finishes the
500 copper.

501 The optimum has two weaknesses. The first is conductivity: 24% filler
502 limits the copper to 64% IACS. The second is robustness. The design is
503 sensitive to the timing on which it rests, and the lower part of Table 3 shows
504 what happens when uncertain inputs are varied one at a time. If the char
505 gasifies half as fast as assumed, the carbon at closure rises from 0.2 to 76 ppm
506 and the peak mismatch from 3.7 to 11.1%; at a quarter of the assumed rate

507 the carbon limit is exceeded by an order of magnitude. A glass transition
 508 15 K higher or lower than calibrated raises the peak mismatch to about 10%.
 509 So does any change in the carbon inhibition scale of copper sintering, C_{inh} ,
 510 by a factor of four or five either way, and at 1000 ppm the damage index of a
 511 buried line exceeds 1. A factor of two in the viscosity of copper changes the
 512 mismatch less (5.8–6.2%), but halving it also takes the damage index above 1.
 513 A design that relies on the timing of char removal is therefore only as good
 514 as the gasification kinetics, the viscosity of the glass and the effect of carbon
 515 on copper, and none of them has been measured for this material pair.

516 The robust search (Section 2.6) therefore asked for a design that meets
 517 every constraint at two corners of the uncertainty in gasification rate and
 518 glass transition and, in addition, keeps the copper above 80% IACS. The
 519 result, listed in the last column of Table 3 and shown in Fig. 6c, follows a
 520 different strategy. It uses a coarser 7 μm copper with almost no filler and a
 521 higher solids loading, and burns out for 2.8 h at 806 $^{\circ}\text{C}$ in 83% steam, some
 522 40 K below the temperature at which the glass would seal during the hold. It
 523 then heats quickly to a short peak at 1017 $^{\circ}\text{C}$ and needs 16.0 h in all. Both
 524 materials start to densify during the burnout hold, the copper once its char
 525 has gone and the glass because it is already soft. Being coarse, the copper
 526 runs only a few per cent ahead of the glass; the glass overtakes it during
 527 the heating to the peak. At nominal inputs the peak mismatch is 9.3%, the
 528 conductivity 89% IACS and the damage index 0.30. In both search corners
 529 the carbon stays below 50 ppm, the damage index below 0.43 and the peak
 530 mismatch at or below 11.4%; the two other corners, not used in the search,
 531 give 9.1 and 9.7% with 34 ppm of carbon at most. Halving the gasification

532 rate or shifting the glass transition by 15 K moves the peak mismatch only
 533 between 9.4 and 9.9%, and the design is equally insensitive to the carbon
 534 inhibition of copper and to the hydrogen inhibition of gasification (9.0–9.6%).
 535 Doubling the viscosity of copper is its weak point: the copper density falls
 536 to 0.912, just below the limit. The robustness is paid for in mismatch. The
 537 nominal value is two and a half times that of the nominal optimum, and
 538 the final camber, -187 m^{-1} , is larger. Most of the remaining mismatch is a
 539 difference in final shrinkage, 8.4%, because the paste now carries more solids
 540 than the glass slurry (56 against 45 vol%), which helps the copper reach
 541 density without filler. A gasification rate four times slower than assumed
 542 defeats both designs, which then leave 1295 and 940 ppm of carbon in the
 543 glass.

544 *4.8. The printed test vehicle*

545 The three designs were then fired in three dimensions as the printed test
 546 vehicle of Fig. 1a: a $12 \times 12 \times 2.4 \text{ mm}$ glass-ceramic block with a buried
 547 $9 \times 9 \text{ mm}$ copper ground plane, a 3.5-turn square spiral with 0.6 mm wide
 548 tracks, and a via to a 1.2 mm pad on the upper face. The block is meshed with
 549 12 800 voxels of 0.3 mm and rests on a setter. The voxel is much coarser than
 550 the pixel of the printer, and every conductor is one voxel, 0.3 mm, thick: three
 551 times the copper of the reference bilayer, which exaggerates the distortion.
 552 Each run starts after binder removal and ends when the glass-ceramic is 98%
 553 dense, at 903–917 °C (Fig. 8). The setter keeps the lower face flat. The
 554 distortion therefore appears on the upper face, which dishes over the buried
 555 plane.

556 The buried plane appears to control the result, although no run without

Table 3: Baseline and optimised designs. Lower part: the two optimised designs re-evaluated with one input changed from its nominal value: the char gasification half-life $t_{1/2}$ (2 h), the glass transition T_g (734 °C), the copper viscosity factor f_η (0.28), the carbon inhibition scale of copper sintering C_{inh} (200 ppm) and the hydrogen inhibition constant of gasification K_H (25). Bold: carbon above 100 ppm; [†]line damage index above 1; [‡]copper density below 0.92.

	Baseline	Nominal optimum	Robust design
<i>Paste and programme</i>			
Copper d_{50} (µm) / filler (vol%) / solids (vol%)	3.0 / 0 / 50	3.9 / 24 / 47	7.0 / 2 / 56
Burnout: T (°C) / time (h) / steam (%)	780 / 4.0 / 30	840 / 1.3 / 81	806 / 2.8 / 83
Matching hold: T (°C) / time (h)	–	916 / 1.7	889 / 0.18
Peak: T (°C) / time (h)	960 / 1.0	1018 / 1.2	1017 / 0.15
Cycle time (h)	17.7	17.3	16.0
<i>Outcome at nominal inputs</i>			
Carbon at glass pore closure (ppm)	33	0.2	0.03
Glass-ceramic density / crystallised fraction	0.991 / 1.00	0.990 / 1.00	0.990 / 1.00
Copper density / conductivity (% IACS)	0.976 / 96	0.965 / 64	0.950 / 89
Peak free-strain mismatch (%)	15.6	3.7	9.3
Line damage index $\max(\Pi_{Cu}, \Pi_{GC})$	1.63	0.83	0.30
Final camber of reference bilayer (m ⁻¹)	–254	–76	–187
<i>Robustness: carbon at closure (ppm) / peak mismatch (%)</i>			
$t_{1/2} = 4$ h	–	76 / 11.1	37 / 9.9
$t_{1/2} = 8$ h	–	1295 / 17.9	940 / 14.4 [‡]
$T_g - 15$ K	–	84 / 9.9	0.06 / 9.4
$T_g + 15$ K	–	0.06 / 9.5	0.03 / 9.8
$f_\eta \times 2$	–	0.2 / 5.8	0.03 / 11.4 [‡]
$f_\eta / 2$	–	0.2 / 6.2 [†]	0.03 / 8.1
$C_{inh} = 50$ ppm	–	0.2 / 9.6	0.03 / 9.5
$C_{inh} = 1000$ ppm	–	0.2 / 9.8 [†]	0.03 / 9.1
$K_H = 5$	–	22 / 8.6	8 / 9.6
$K_H = 100$	32	0.0 / 10.2	0.0 / 9.0

557 it was made. In the baseline the copper has already densified when the glass
 558 begins to flow, so the plane acts as a rigid inclusion. The block shrinks by
 559 only 14.9% in plane, against 23.1% for the free glass-ceramic, and the glass
 560 above the plane, unable to shrink laterally, densifies by shrinking more in
 561 thickness. The upper face ends 622 μm lower at the centre than at the rim
 562 (Fig. 8c), about a third of the fired thickness. In the nominal optimum the
 563 copper and the glass densify together. The plane then offers little resistance,
 564 the in-plane shrinkage rises to 21.2%, and the relief of the upper face falls to
 565 150 μm . The robust design lies between them: 18.3% in-plane shrinkage and
 566 317 μm of relief. The relief follows the peak mismatch of the one-dimensional
 567 analysis rather than the final one: the robust design ends with a larger final
 568 mismatch than the baseline (8.4 against 4.0%) but with half its relief. The
 569 via and pad raise a small plateau at the centre of every part. That the three
 570 designs rank in the same order by relief and by peak mismatch supports the
 571 use of the one-dimensional analysis for design. The absolute values, however,
 572 depend on the area of copper. A plane that covers 56% of the part, as here,
 573 is the most demanding case, and a real layout would divide it or add dummy
 574 metal in the rim.

575 *4.9. Cooling from the stress-free temperature*

576 Below about 690 $^{\circ}\text{C}$, the strain point of the calibrated glass, mismatch
 577 stresses no longer relax. The difference in expansion between copper (16.5–
 578 22 ppm K^{-1} over this range) and the glass-ceramic (3 ppm K^{-1}) must then
 579 be carried by the solids. Fig. 9 shows the consequences for a copper plane
 580 on 1 mm of glass-ceramic. If copper remained elastic, its mean stress would
 581 exceed 900 MPa at room temperature and a 20 mm stack with 100 μm of

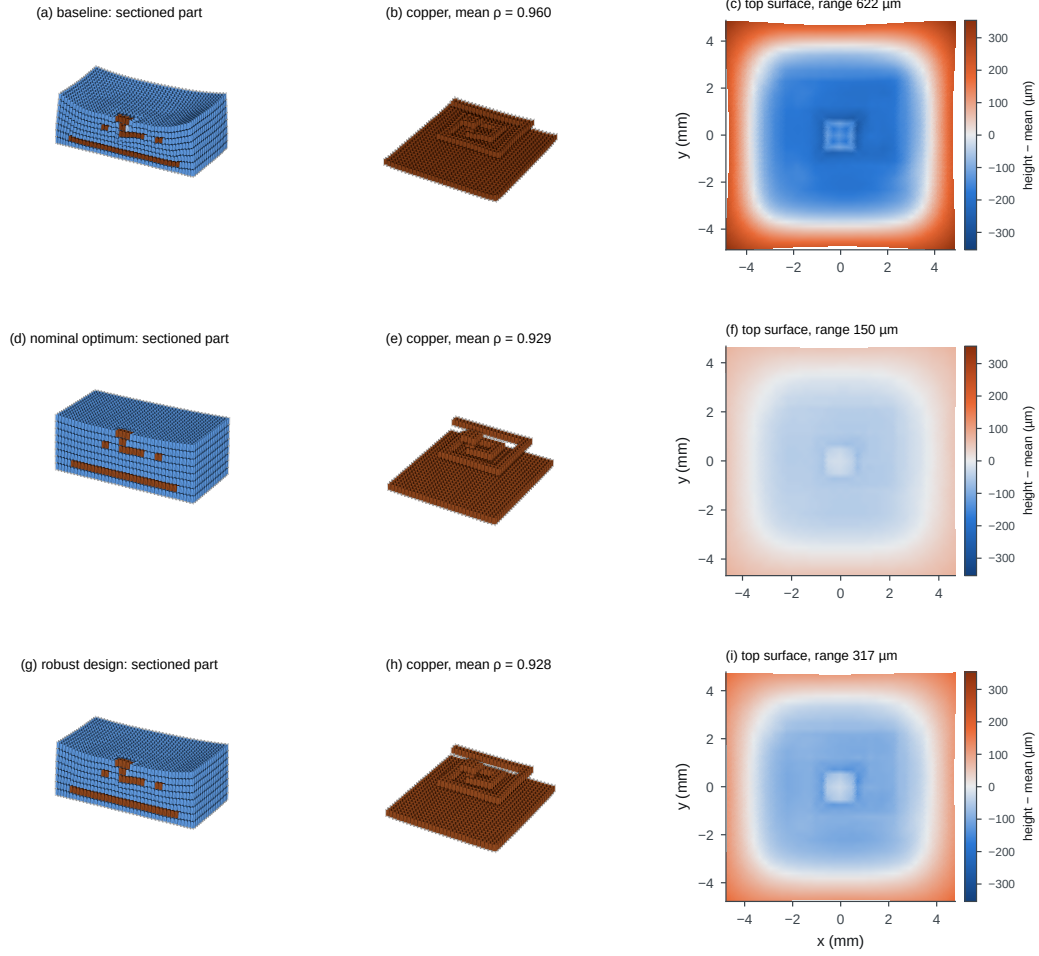


Figure 8: Three-dimensional co-sintering of the printed test vehicle (0.3 mm voxels, setter support, gravity) for the three designs of Table 3, from binder removal to 98% density of the glass-ceramic. Left: part sectioned through its centre, showing the buried plane, the spiral and the via. Centre: the copper, coloured by density. Right: height of the upper face relative to its mean; the lower face is held flat by the setter.

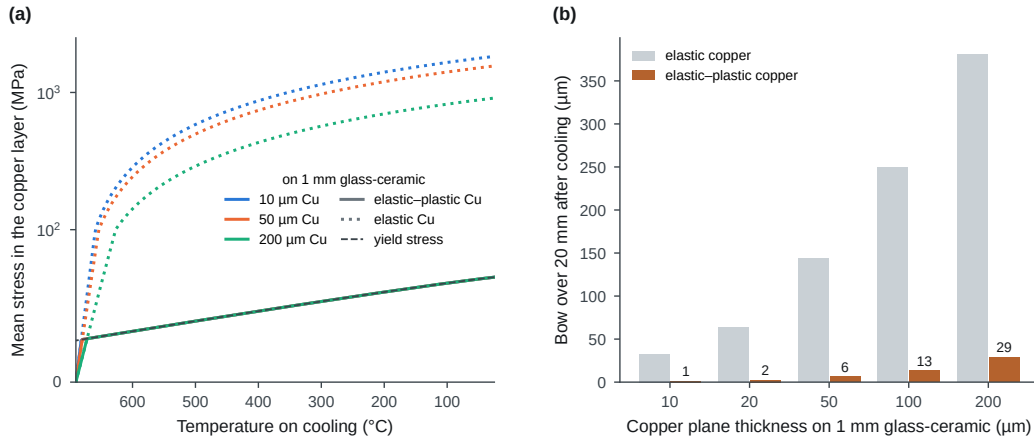


Figure 9: Cooling of a copper plane on 1 mm of glass-ceramic from the stress-free temperature (690 °C). (a) Mean stress in the copper layer for elastic and elastic-perfectly plastic copper; the plastic curves coincide with the temperature-dependent yield stress. (b) Bow over a 20 mm span at room temperature.

582 copper would bow by 250 μm. Annealed copper cannot carry that stress. It
583 yields within about 13 K of cooling and then follows its yield stress, which rises
584 from about 27 MPa at 690 °C to 69 MPa at room temperature. The bow falls
585 accordingly, to 13 μm for 100 μm of copper and to 6 μm for 50 μm, and the
586 glass-ceramic stays in mild compression (−7 MPa for 100 μm of copper). The
587 result does not depend on the stress-free temperature: starting the analysis
588 at 500, 600, 700 or 800 °C gives the same final stress and bow, because the
589 copper yields almost at once in every case. Plastic accommodation makes
590 the thermal-expansion mismatch a secondary problem compared with the
591 sintering mismatch, provided the copper is dense and ductile. The price is
592 cyclic plasticity in the conductor, which matters for thermal cycling in service
593 but is outside the scope of this study.

594 4.10. Which inputs decide the outcome

595 Fig. 10 ranks the fourteen uncertain inputs by their elementary effects
596 around the nominal optimum. Elementary effects are expressed per full
597 range of each factor. The half-life of char gasification dominates: across
598 its range of 0.5–8 h it changes the peak mismatch by 16 percentage points
599 on average and the carbon at glass closure by several orders of magnitude,
600 from well below 1 ppm to above the 100 ppm limit. It also comes second
601 for copper density, because char that is not removed keeps the copper from
602 sintering. The carbon trapped in the glass depends next on the fragility and
603 transition temperature of the glass, an order of magnitude less strongly. The
604 density of the glass-ceramic is governed by its own properties: the transition
605 temperature, the particle size and the crystallisation kinetics. The density
606 and conductivity of the copper depend on the viscosity factor f_η and on the
607 filler. The peak mismatch, the quantity that the design sought to minimise,
608 responds to almost every input after the gasification rate: f_η , the glass
609 transition, the char yield, and the particle size and solids loading of the glass.
610 For every output that matters, σ is of the same order as μ^* . The effects are
611 therefore strongly non-linear and interact, which is what a design based on
612 the coincidence of two processes should be expected to show. The tolerances
613 on the copper paste itself (particle size -25 to $+33\%$, filler ± 5 vol%) matter
614 for the damage index of a buried line, together with f_η and the char yield,
615 and the filler tolerance also for the conductivity. The Avrami exponent and
616 the activation energy of crystallisation matter only for the density of the
617 glass-ceramic, and the crystallised fraction (not shown) remained complete
618 in every run. Three inputs were left out of the screening and checked one

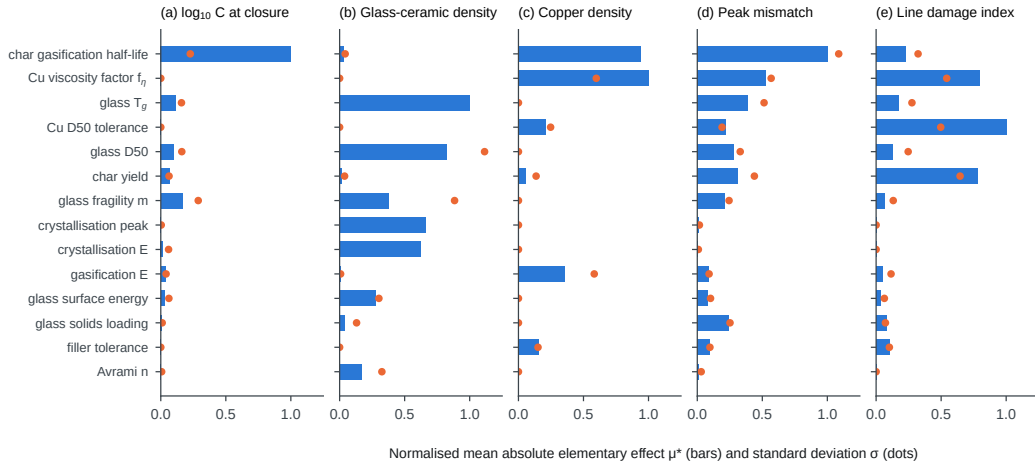


Figure 10: Morris screening of the fourteen uncertain inputs around the nominal optimum (16 trajectories, 240 co-firing evaluations). Bars: mean absolute elementary effect μ^* ; dots: standard deviation σ of the elementary effects, both normalised by the largest μ^* of each output. Factors are ordered by their summed normalised importance. Ranges as in Table 1; the copper particle size and filler fraction are varied as manufacturing tolerances around the design.

at a time instead (Table 3): the carbon inhibition scale of copper sintering, the hydrogen inhibition of gasification and the density at which the glass closes its pores. The last has little effect; the peak mismatch of the nominal optimum stays between 3.5 and 3.9%. The first two matter as much as the gasification rate for the nominal optimum, and hardly at all for the robust design.

5. Discussion

5.1. The order of events in co-firing

Taken together, the results fix an order of events that any firing programme for this material pair has to respect. Pyrolysis in nitrogen leaves about 2%

629 of char, relative to the solids, in the glass-ceramic and about 0.5% in the
 630 copper. The char must then be gasified before the glass seals its pores. With
 631 the atmosphere held at a fixed fraction of the copper-oxide equilibrium, this
 632 requires hours at 770–800 °C and is the slowest step of the process. In the
 633 model, removing the char also releases the copper. Residual carbon is assumed
 634 to retard the sintering of copper (Eq. (7)), so a fine powder that has been
 635 held back by its own char starts to densify as soon as the steam clears it, at
 636 680–780 °C in the baseline and well before the glass. This effect has not been
 637 measured for photopolymer char, and its strength, C_{inh} , is an estimate. In
 638 silver-metallised LTCC fired in air this coupling does not arise, because the
 639 binder burns off completely at low temperature and the metal is free to sinter
 640 from then on. Here the burnout step is also, whether intended or not, the
 641 first sintering step of the conductor. The glass then densifies over a range
 642 of about 80 K and stops when it crystallises, and below the strain point of
 643 the residual glass the expansion mismatch is taken up by plastic flow of the
 644 copper.

645 This order explains why neither lever works alone. The furnace programme
 646 cannot delay the densification of fine copper beyond the burnout, because the
 647 burnout has to happen below the coalescence of the glass. The paste cannot
 648 be made as late and as steep as the glass (Fig. 5) without losing conductor
 649 density. The two designs of Table 3 take different approaches. The nominal
 650 optimum uses the assumed coupling between char and copper to start both
 651 materials together. It reaches the lowest mismatch, but it depends on the
 652 timing of char removal and on the strength of the coupling, which are among
 653 the least known inputs of the model. The robust design gives up part of that

654 gain and relies on a coarse copper that is intrinsically slow. It tolerates a
655 factor of two in the gasification rate together with 15 K in the glass transition,
656 and it is insensitive to the assumed effect of carbon on copper. Which of the
657 two is preferable depends on what has been measured. Once the gasification
658 kinetics, the viscosity of the glass and the effect of carbon on copper are
659 known much more tightly than these bounds, a design closer to the nominal
660 optimum becomes safe.

661 *5.2. Relation to tape-based co-firing*

662 The rule used for the atmosphere, hydrogen tied to steam so that the
663 water-to-hydrogen ratio sits at a fixed fraction of the copper limit, is consistent
664 with the operating point of the IBM process (Fig. 3), and the calibrated glass
665 is consistent with the burnout temperature chosen for it (Table 2). The model
666 adds no new principle to that process, but it makes it quantitative: how long
667 the burnout must last at a given temperature and steam content, how much
668 carbon is left when the pores close, and how the choice of copper powder
669 interacts with the programme. In LTCC, camber is controlled by adjusting
670 the paste formulation [39] or, with master-sintering-curve predictions, the
671 heating schedule [40]. For copper in a crystallising glass, Fig. 5 shows why
672 paste formulation alone is not sufficient. The heating programme sets the
673 order in which the materials are released to sinter, and the paste then adjusts
674 the rate.

675 Printing changes the problem in three respects. First, a photocurable
676 slurry carries more binder than a cast tape: at 45 vol% solids the glass-ceramic
677 shrinks by 23% linearly, against 12.7–15.8% in the plane of commercial LTCC
678 tapes [81, 82], and it holds correspondingly more char. Second, conductors

are no longer confined to printed planes connected by punched vias. The test vehicle contains a spiral, a buried plane and a via defined in one build. In the three-dimensional model the large buried plane appears to control the distortion; tape technology would mesh or perforate such a plane, and a printed layout can divide it freely. Third, printed ceramics may shrink anisotropically between the build direction and the plane, as laminated tapes do: unconstrained IBM laminates fired in H_2/H_2O shrink about half as much through the thickness as in the plane [79]. The model treats both materials as isotropic. Anisotropy would change the camber and the shape of the fired part, but not the order of events described above.

5.3. Limitations

The results rest on published data and a model; no part has been fired. Four inputs deserve particular caution. The viscosity of the glass was calibrated against statements in patents, not measured, and it describes one glass composition; a different cordierite-type glass will shift the whole window. The gasification kinetics of photopolymer char on either material are unknown and were estimated, so the absolute times of the burnout carry the largest uncertainty of the study (Section 4.10). The retarding effect of carbon on copper sintering is an assumption of the model whose strength was estimated. The viscosity factor of copper was fitted to a coarser powder than those considered here. In addition, the model omits several phenomena that can matter in practice: dissolution of copper oxide in the glass and interdiffusion at the interface, adhesion and delamination, the water released by the glass during crystallisation, the effect of residual carbon on dielectric loss, and cyclic plasticity of the copper in service. The damage indices compare stresses

704 with strength estimates and are meant for ranking designs, not for predicting
705 failure. The three-dimensional runs stop at 98% density of the glass-ceramic,
706 so the final densification of the copper at the peak temperature, inside an
707 already rigid body, is represented only in the slab model.

708 5.4. *What to measure first*

709 The sensitivity analysis (Fig. 10) and the one-at-a-time checks of Table 3
710 turn the limitations into a short list of measurements, ordered by their value
711 to the design:

- 712 1. The gasification kinetics of the photopolymer char on each material, in
713 the steam–hydrogen atmosphere of the burnout. Interrupted burnout runs
714 followed by carbon analysis at three temperatures would fix both the rate
715 and its activation energy. No other quantity has as large an effect on the
716 carbon trapped in the glass and on the mismatch.
- 717 2. The viscosity of the glass between 750 and 900 °C, which fixes T_g and
718 m . Beam bending or parallel-plate viscometry of the glass would give it
719 directly, and so would isothermal dilatometry of the printed glass-ceramic,
720 interpreted with the closed form of Appendix B.
- 721 3. The sintering viscosity of the copper slurry, f_η , from dilatometry of printed
722 copper coupons in dry hydrogen after binder removal.
- 723 4. The char yield of both cured slurries, from thermogravimetry in nitrogen.
- 724 5. Whether, and how strongly, residual char slows the sintering of the copper
725 slurry, from dilatometry of copper coupons after partial and complete
726 burnout. The nominal optimum depends on it; the robust design does not.
- 727 The particle size and solids loading of the glass also matter, but they are
728 specified by the supplier and the slurry formulation. The crystallisation

729 kinetics matter little within the ranges considered, as long as the exotherm
730 peak lies above the sintering range. All of these measurements are routine,
731 and most of them can be made on single-material coupons printed with the
732 same slurries.

733 *5.5. Implications for three-dimensional packages*

734 A fired body of this kind would have the properties that made glass-
735 ceramic/copper attractive for high-end packaging. Its thermal expansion,
736 3.0 ppm K^{-1} , is close to that of silicon (2.6 ppm K^{-1}) [71], its permittivity is
737 about 5 [13], and its conductors are copper. From porosity and filler alone
738 the model predicts 89% IACS for the conductor of the robust design and 96%
739 for a paste without filler. IBM reported $2.0 \pm 0.2 \text{ } \mu\Omega \text{ cm}$, about 86% IACS,
740 for copper co-fired in $\text{H}_2/\text{H}_2\text{O}$ [13]. The difference points to impurities and
741 grain-boundary oxygen, which the model does not include. Printing adds the
742 geometric freedom that tape lamination lacks. Vertical coils, shielded lines,
743 cavities and ground planes of arbitrary shape can be built into one part. That
744 freedom is relevant to antenna-in-package modules and millimetre-wave front
745 ends [3, 2], and to substrates for wide-bandgap power devices [4]. The present
746 study defines a candidate firing route for such parts and the few quantities
747 that have to be measured before it is tried.

748 **6. Conclusions**

749 A model of binder removal, sintering and crystallisation was used to design
750 the co-firing of copper inside a cordierite-type glass-ceramic printed as one
751 part by multi-material vat photopolymerisation. It was verified against closed-
752 form solutions, and its glass was calibrated against the published IBM glass-

ceramic/copper process, with whose atmosphere and burnout temperature it is consistent. No experiments were made. The main conclusions are as follows.

1. Of the atmospheres examined, only steam removes the carbon from the glass before its pores close. Hydrogen tied to the steam at a fixed fraction of the Cu/Cu₂O equilibrium keeps the copper metallic and is consistent with the IBM operating point. At 780 °C such an atmosphere gasifies char seven times faster than humidified forming gas, which leaves more than 300 ppm of carbon in the glass-ceramic in every programme examined.
2. Fine copper and the glass densify in different temperature ranges: 450–700 °C for 3 µm copper against 850–930 °C for the glass. Co-fired with the IBM-type programme, the copper densifies during the burnout, the free-strain mismatch reaches 15.6%, and the damage index of a buried line exceeds 1. The final shrinkages differ by only 4%; the difficulty lies in the order in which the two materials shrink.
3. If char holds copper back, as the model assumes, the burnout step also releases the copper to sinter. The best paste and programme found exploit this coupling: a short, hot burnout just below the coalescence of the glass releases both materials together and reduces the peak mismatch to 3.7%. The design is sensitive to this timing. Halving the gasification rate, shifting the glass transition by 15 K or changing the strength of the carbon effect by a factor of four to five restores one half to two thirds of the mismatch that the optimisation removed, and the 24% filler limits the copper to 64% IACS.
4. A design required to meet every constraint under a factor of two either

way in the gasification rate combined with ± 15 K in the glass transition
 uses 7 μm copper with almost no filler and a 2.8 h burnout at 806 °C. At
 all four corners of that uncertainty it keeps the peak mismatch between
 9.1 and 11.4%, the copper at 89% IACS and the carbon at glass closure
 below 50 ppm, in a 16 h cycle, and it is insensitive to the assumed effect
 of carbon on copper. The robustness costs a factor of two and a half in
 nominal mismatch.

5. In three dimensions, a buried copper plane that densifies before the glass
 acts as a rigid inclusion. The baseline test vehicle shrinks by 15% in plane
 instead of 23%, and its upper face dishes by 0.6 mm. The nominal optimum
 reduces the dishing to 0.15 mm and the robust design to 0.32 mm, the
 same ranking as the one-dimensional mismatch.
6. On cooling, plastic flow of the copper limits the stress it transmits to its
 yield stress. A 100 μm copper plane on 1 mm of glass-ceramic bows by
 13 μm over 20 mm, about a nineteenth of the elastic prediction.
7. The inputs that decide the outcome are, in order, the gasification kinetics
 of the char, the viscosity of the glass, the sintering viscosity of the copper
 and the char yield. All four can be measured on single-material printed
 coupons, and they should be measured before the process is tried.

CRediT authorship contribution statement

[Author names]: Conceptualization, Methodology, Software, Formal
 analysis, Visualization, Writing – original draft, Writing – review & editing.

800 **Declaration of competing interest**

801 The authors declare that they have no known competing financial interests
802 or personal relationships that could have appeared to influence the work
803 reported in this paper.

804 **Data availability**

805 The simulation code, input parameters with their provenance, and the
806 scripts that generate every figure in this paper are available from the corre-
807 sponding author on reasonable request.

808 **Declaration of generative AI and AI-assisted technologies in the** 809 **writing process**

810 During the preparation of this work the authors used [name of tool] in
811 order to [reason, e.g. assist with code development, data analysis and drafting
812 of the text]. After using this tool, the authors reviewed and edited the content
813 as needed and take full responsibility for the content of the publication.

814 **References**

- 815 [1] J. H. Lau, Recent advances and trends in advanced packaging, IEEE
816 Transactions on Components, Packaging and Manufacturing Technology
817 12 (2) (2022) 228–252. doi:10.1109/tcpmt.2022.3144461.
- 818 [2] A. O. Watanabe, M. Ali, S. Y. B. Sayeed, R. R. Tummala, M. R.
819 Pulugurtha, A review of 5g front-end systems package integration, IEEE

- 820 Transactions on Components, Packaging and Manufacturing Technology
821 11 (1) (2021) 118–133. doi:10.1109/tcpmt.2020.3041412.
- 822 [3] Y. Zhang, J. Mao, An overview of the development of antenna-in-package
823 technology for highly integrated wireless devices, Proceedings of the IEEE
824 107 (11) (2019) 2265–2280. doi:10.1109/jproc.2019.2933267.
- 825 [4] C. Chen, F. Luo, Y. Kang, A review of sic power module packaging:
826 Layout, material system and integration, CPSS Transactions on Power
827 Electronics and Applications 2 (3) (2017) 170–186. doi:10.24295/
828 cpsstpea.2017.00017.
- 829 [5] M. T. Sebastian, H. Jantunen, Low loss dielectric materials for ltcc
830 applications: a review, International Materials Reviews 53 (2) (2008)
831 57–90. doi:10.1179/174328008x277524.
- 832 [6] R. R. Tummala, Ceramic and glass-ceramic packaging in the 1990s,
833 Journal of the American Ceramic Society 74 (5) (1991) 895–908. doi:
834 10.1111/j.1151-2916.1991.tb04320.x.
- 835 [7] Y. Imanaka, Multilayered Low Temperature Cofired Ceramics (LTCC)
836 Technology, Kluwer Academic Publishers (Springer), 2005. doi:10.
837 1007/b101196.
- 838 [8] IEEE Electronics Packaging Society, Heterogeneous integration roadmap,
839 2024 edition, chapter 25: Additive manufacturing & additive electronics
840 for heterogeneous integration, Tech. rep., IEEE Electronics Packaging So-
841 ciety, chapter 25, [https://eps.ieee.org/wp-content/uploads/2025/](https://eps.ieee.org/wp-content/uploads/2025/11/HIR_2024_ch25_AME.pdf)
842 11/HIR_2024_ch25_AME.pdf (May 2024).

- [9] S. Yang, A. Christou, Failure model for silver electrochemical migration, IEEE Transactions on Device and Materials Reliability 7 (1) (2007) 188–196. doi:10.1109/tdmr.2007.891531.
- [10] J.-H. Jean, C.-R. Chang, Interfacial reaction kinetics between silver and ceramic-filled glass substrate, Journal of the American Ceramic Society 87 (7) (2004) 1287–1293. doi:10.1111/j.1151-2916.2004.tb07724.x.
- [11] M. Ma, Z. Liu, F. Zhang, F. Liu, Y. Li, Suppression of silver diffusion in borosilicate glass-based low-temperature cofired ceramics by copper oxide addition, Journal of the American Ceramic Society 99 (7) (2016) 2402–2407. doi:10.1111/jace.14248.
- [12] J. A. Lewis, Binder removal from ceramics, Annual Review of Materials Science 27 (1) (1997) 147–173. doi:10.1146/annurev.matsci.27.1.147.
- [13] L. W. Herron, R. N. Master, R. R. Tummala, Method of making multilayered glass-ceramic structures having an internal distribution of copper-based conductors, US Patent 4,234,367, issued 1980-11-18, assigned to IBM (1980).
- [14] R. Master, L. Herron, R. Tummala, Cosintering process for glass-ceramic/copper multilayer ceramic substrate, IEEE Transactions on Components, Hybrids, and Manufacturing Technology 14 (4) (1991) 780–783. doi:10.1109/33.105133.
- [15] M. Schwentenwein, J. Homa, Additive manufacturing of dense alumina

- ceramics, *International Journal of Applied Ceramic Technology* 12 (1) (2015) 1–7. doi:10.1111/ijac.12319.
- [16] J. Stampfl, M. Schwentenwein, J. Homa, F. B. Prinz, Lithography-based additive manufacturing of ceramics: Materials, applications and perspectives, *MRS Communications* 13 (5) (2023) 786–794. doi:10.1557/s43579-023-00444-0.
- [17] J. W. Halloran, Ceramic stereolithography: Additive manufacturing for ceramics by photopolymerization, *Annual Review of Materials Research* 46 (1) (2016) 19–40. doi:10.1146/annurev-matsci-070115-031841.
- [18] L. Qian, E. Hayward, M. Salek, Z. Liu, J. Vihinen, Y. Wang, 3-d printed monolithic dielectric waveguide filter using lcm technique, in: 2022 IEEE MTT-S International Microwave Workshop Series on Advanced Materials and Processes for RF and THz Applications (IMWS-AMP), IEEE, 2022, pp. 1–3. doi:10.1109/imws-amp54652.2022.10106895.
- [19] A. Fontana, A. Delage, A. Périgaud, P. Richard, D. Carsenat, G. Acikalin, S. Verdeyme, B. Bonnet, L. Carpentier, N. Delhote, A novel approach toward the integration of fully 3-d printed surface-mounted microwave ceramic filters, *IEEE Transactions on Microwave Theory and Techniques* 71 (9) (2023) 3915–3928. doi:10.1109/tmtt.2023.3267541.
- [20] G. Mazingue, M. Romier, N. Capet, 3d printed ceramic low-profile gnss antenna for smallsats, in: 2020 50th European Microwave Conference (EuMC), IEEE, 2021, pp. 460–462. doi:10.23919/eumc48046.2021.9337981.

- 888 [21] J. Fernandes, P. Barcelona, M. Blanes, J. Padilla, F. Ramos, A. Cir-
889 era, E. Xuriguera, Study of mixing process of low temperature co-
890 fired ceramics photocurable suspension for digital light processing
891 stereolithography, *Ceramics International* 47 (11) (2021) 15931–15938.
892 doi:10.1016/j.ceramint.2021.02.167.
- 893 [22] P. Wang, X. Chen, H. Zhang, Z. Chen, Y. Cong, B. Xu, M. Chen,
894 Y. Zhang, J. Li, Fabrication of high-resolution 3d ceramic electronics via
895 in situ laser-activated selective electroless plating, *Advanced Materials*
896 38 (48) (2026) e74140. doi:10.1002/adma.74140.
- 897 [23] T. Hirao, S. Hamada, Novel multi-material 3-dimensional low-
898 temperature co-fired ceramic base, *IEEE Access* 7 (2019) 12959–12963.
899 doi:10.1109/access.2019.2892654.
- 900 [24] J. Jäger, M. Ihle, K. Gläser, A. Zimmermann, Inkjet-printed low temper-
901 ature co-fired ceramics: process development for customized ltcc, *Flexible*
902 *and Printed Electronics* 9 (2) (2024) 025022. doi:10.1088/2058-8585/
903 ad59b3.
- 904 [25] C. Liang, J. Huang, J. Wang, H. Gong, D. Bai, P. Zhao, Additive
905 manufacturing of low-temperature co-fired ceramic substrates and surface
906 conductors based on material jetting, *Additive Manufacturing* 78 (2023)
907 103856. doi:10.1016/j.addma.2023.103856.
- 908 [26] P. Duan, X. Zhu, H. Zhang, H. Li, Z. Li, R. Wang, J. Zhou, D. Song,
909 Y. Zhang, G. Zhang, H. Lan, Cost-effective fabrication of customized
910 ltcc devices with multilayer using multi-material 3d printing, *Journal*

- 911 of Manufacturing Processes 107 (2023) 88–97. doi:10.1016/j.jmapro.
912 2023.10.043.
- 913 [27] S. Geier, I. Potestio, 3D-printing: From multi-material to
914 functionally-graded ceramic, Ceramic Applications 8 (2) (2020) 32–35,
915 [https://lithoz.com/wp-content/uploads/2022/06/3Dprinting_-_](https://lithoz.com/wp-content/uploads/2022/06/3Dprinting_-_from_multi-material_to_fuctionally-graded_ceramic.pdf)
916 [from_multi-material_to_fuctionally-graded_ceramic.pdf](https://lithoz.com/wp-content/uploads/2022/06/3Dprinting_-_from_multi-material_to_fuctionally-graded_ceramic.pdf).
- 917 [28] S. Xu, C. Huang, H. Liu, J. Huang, Vat photopolymerization additive
918 manufacturing for advanced ceramics: Techniques, multimaterial strate-
919 gies, and applications, Advanced Engineering Materials 28 (5) (2026)
920 e202501811. doi:10.1002/adem.202501811.
- 921 [29] S. Subedi, S. Liu, W. Wang, S. M. A. Naser Shovon, X. Chen, H. O. T.
922 Ware, Multi-material vat photopolymerization 3d printing: a review of
923 mechanisms and applications, npj Advanced Manufacturing 1 (1) (2024)
924 9. doi:10.1038/s44334-024-00005-w.
- 925 [30] J. Schlacher, S. Geier, M. Schwentenwein, R. Bermejo, Towards 3d-
926 printed alumina-based multi-material components with enhanced thermal
927 shock resistance, Journal of the European Ceramic Society 44 (4) (2024)
928 2294–2303. doi:10.1016/j.jeurceramsoc.2023.11.009.
- 929 [31] E. Schwarzer-Fischer, A. Günther, S. Roszeitis, T. Moritz, Combining
930 zirconia and titanium suboxides by vat photopolymerization, Materials
931 14 (9) (2021) 2394. doi:10.3390/ma14092394.
- 932 [32] M. Roumanie, C. Flassayer, A. Resch, L. Cortella, R. Laucournet, In-
933 fluence of debinding and sintering conditions on the composition and

- 934 thermal conductivity of copper parts printed from highly loaded pho-
 935 tocurable formulations, SN Applied Sciences 3 (1) (2021) 55. doi:
 936 10.1007/s42452-020-04049-3.
- 937 [33] J. Scheibler, A. S. Kosmehl, T. Studnitzky, C. Zhong, T. Weißgärber,
 938 Lithography-based metal manufacturing of copper: Influence of exposure
 939 parameters on green part strength, Metals 14 (11) (2024) 1268. doi:
 940 10.3390/met14111268.
- 941 [34] A. Resch, A. Benayad, M. Roumanie, C. Croutxé-Barghorn, Lithography
 942 based metal manufacturing (lmm): Influence of particle size and compo-
 943 sition of copper powder on uv light penetration, Materials Today Com-
 944 munications 35 (2023) 105595. doi:10.1016/j.mtcomm.2023.105595.
- 945 [35] S. Cano Cano, P. Peritsch, J. Bosters, A. Anand, C. Gierl-Mayer,
 946 G. Attila Harakály, Recent advances of lithography-based metal man-
 947 ufacturing of copper, in: Euro PM2024 Proceedings, EPMA, 2024.
 948 doi:10.59499/ep246281350.
- 949 [36] P. Z. Cai, D. J. Green, G. L. Messing, Constrained densification of
 950 alumina/zirconia hybrid laminates, i: Experimental observations of
 951 processing defects, Journal of the American Ceramic Society 80 (8)
 952 (1997) 1929–1939. doi:10.1111/j.1151-2916.1997.tb03075.x.
- 953 [37] D. J. Green, O. Guillon, J. Rödel, Constrained sintering: A delicate
 954 balance of scales, Journal of the European Ceramic Society 28 (7) (2008)
 955 1451–1466. doi:10.1016/j.jeurceramsoc.2007.12.012.

- 956 [38] J.-H. Jean, C.-R. Chang, Cofiring kinetics and mechanisms of an ag-
957 metallized ceramic-filled glass electronic package, *Journal of the American*
958 *Ceramic Society* 80 (12) (1997) 3084–3092. doi:10.1111/j.1151-2916.
959 1997.tb03236.x.
- 960 [39] C.-R. Chang, J.-H. Jean, Effects of silver-paste formulation on camber
961 development during the cofiring of a silver-based, low-temperature-cofired
962 ceramic package, *Journal of the American Ceramic Society* 81 (11) (1998)
963 2805–2814. doi:10.1111/j.1151-2916.1998.tb02700.x.
- 964 [40] B. Brandt, H. Naghib-zadeh, T. Rabe, Improved co-firing of ferrite and
965 dielectric tape based on master sintering curve predictions and shrinkage
966 mismatch calculations, *Journal of the American Ceramic Society* 96 (3)
967 (2013) 726–730. doi:10.1111/jace.12179.
- 968 [41] E. Olevsky, T. T. Molla, H. L. Frandsen, R. Bjørk, V. Esposito, D. W.
969 Ni, A. Ilyina, N. Pryds, Sintering of multilayered porous structures: Part
970 i-constitutive models, *Journal of the American Ceramic Society* 96 (8)
971 (2013) 2657–2665. doi:10.1111/jace.12375.
- 972 [42] T. T. Molla, D. W. Ni, R. Bulatova, R. Bjørk, C. Bahl, N. Pryds, H. L.
973 Frandsen, Finite element modeling of camber evolution during sintering
974 of bilayer structures, *Journal of the American Ceramic Society* 97 (9)
975 (2014) 2965–2972. doi:10.1111/jace.13025.
- 976 [43] J. Kanters, U. Eisele, J. Rödel, Cosintering simulation and experimenta-
977 tion: Case study of nanocrystalline zirconia, *Journal of the American*

- 978 Ceramic Society 84 (12) (2001) 2757–2763. doi:10.1111/j.1151-2916.
979 2001.tb01091.x.
- 980 [44] M. W. Reiterer, K. G. Ewsuk, J. G. Argüello, An arrhenius-type vis-
981 cosity function to model sintering using the skorohod–olevsky viscous
982 sintering model within a finite-element code, Journal of the American
983 Ceramic Society 89 (6) (2006) 1930–1935. doi:10.1111/j.1551-2916.
984 2006.01041.x.
- 985 [45] L. Chrétien, A. Heux, G. Antou, N. Pradeilles, N. Delhote, A. Maître,
986 Distortion of an ltcc bilayer during constrained sintering: Comparison
987 between ombroscopic imaging and modeling, Materials 15 (18) (2022)
988 6405. doi:10.3390/ma15186405.
- 989 [46] G. Antou, A. Heux, N. Pradeilles, N. Delhote, A. Maître, Co-sintering of
990 a ceramic-metal bilayer: Coupled experimental, analytical and numerical
991 approaches, Ceramics International 50 (22) (2024) 46196–46210. doi:
992 10.1016/j.ceramint.2024.08.462.
- 993 [47] M. W. Chase, Jr., NIST-JANAF Thermochemical Tables, 4th Edition,
994 Journal of Physical and Chemical Reference Data, Monograph 9, Ameri-
995 can Institute of Physics, Woodbury, NY, 1998.
- 996 [48] E. A. Olevsky, Theory of sintering: from discrete to continuum, Materials
997 Science and Engineering: R: Reports 23 (2) (1998) 41–100. doi:10.1016/
998 S0927-796X(98)00009-6.
- 999 [49] J. C. Mauro, Y. Yue, A. J. Ellison, P. K. Gupta, D. C. Allan, Viscosity of

- 1000 glass-forming liquids, Proceedings of the National Academy of Sciences
1001 106 (47) (2009) 19780–19784. doi:10.1073/pnas.0911705106.
- 1002 [50] M. Avrami, Kinetics of phase change. i general theory, The Journal of
1003 Chemical Physics 7 (12) (1939) 1103–1112. doi:10.1063/1.1750380.
- 1004 [51] H. E. Kissinger, Reaction kinetics in differential thermal analysis, Analyt-
1005 ical Chemistry 29 (11) (1957) 1702–1706. doi:10.1021/ac60131a045.
- 1006 [52] I. M. Krieger, T. J. Dougherty, A mechanism for non-newtonian flow
1007 in suspensions of rigid spheres, Transactions of the Society of Rheology
1008 3 (1) (1959) 137–152. doi:10.1122/1.548848.
- 1009 [53] H. J. Frost, M. F. Ashby, Deformation-Mechanism Maps: The Plasticity
1010 and Creep of Metals and Ceramics, Pergamon Press, Oxford, 1982.
- 1011 [54] J. C. M. Garnett, Xii. colours in metal glasses and in metallic films,
1012 Philosophical Transactions of the Royal Society of London. Series A,
1013 Containing Papers of a Mathematical or Physical Character 203 (359-371)
1014 (1904) 385–420. doi:10.1098/rsta.1904.0024.
- 1015 [55] S. Timoshenko, Analysis of bi-metal thermostats, Journal of the Optical
1016 Society of America 11 (3) (1925) 233. doi:10.1364/JOSA.11.000233.
- 1017 [56] P. Z. Cai, D. J. Green, G. L. Messing, Constrained densification of
1018 alumina/zirconia hybrid laminates, ii: Viscoelastic stress computation,
1019 Journal of the American Ceramic Society 80 (8) (1997) 1940–1948. doi:
1020 10.1111/j.1151-2916.1997.tb03076.x.

- 1021 [57] J. D. Eshelby, The determination of the elastic field of an ellipsoidal
1022 inclusion, and related problems, Proceedings of the Royal Society of
1023 London. Series A. Mathematical and Physical Sciences 241 (1226) (1957)
1024 376–396. doi:10.1098/rspa.1957.0133.
- 1025 [58] T. Gustafsson, G. McBain, scikit-fem: A python package for finite
1026 element assembly, Journal of Open Source Software 5 (52) (2020) 2369.
1027 doi:10.21105/joss.02369.
- 1028 [59] N. Bell, L. N. Olson, J. Schroder, Pyamg: Algebraic multigrid solvers
1029 in python, Journal of Open Source Software 7 (72) (2022) 4142. doi:
1030 10.21105/joss.04142.
- 1031 [60] L. F. Shampine, M. W. Reichelt, The matlab ode suite, SIAM
1032 Journal on Scientific Computing 18 (1) (1997) 1–22. doi:10.1137/
1033 S1064827594276424.
- 1034 [61] R. Storn, K. Price, Differential evolution – a simple and efficient heuris-
1035 tic for global optimization over continuous spaces, Journal of Global
1036 Optimization 11 (4) (1997) 341–359. doi:10.1023/A:1008202821328.
- 1037 [62] M. D. Morris, Factorial sampling plans for preliminary computational
1038 experiments, Technometrics 33 (2) (1991) 161–174. doi:10.1080/
1039 00401706.1991.10484804.
- 1040 [63] A. H. Kumar, P. W. McMillan, R. R. Tummala, Glass-ceramic structures
1041 and sintered multilayer substrates thereof with circuit patterns of gold,
1042 silver or copper, US Patent 4,301,324, issued 1981-11-17, assigned to
1043 IBM (1981).

- 1044 [64] Y. Yu, J. Wang, Y. Yu, Z. Yan, Y. Du, P. Chu, Q. Jing, P. Liu, Synthesis
1045 and characterization of single-phase α -cordierite glass-ceramics for ltcc
1046 substrates from tuff, *Materials* 15 (24) (2022) 8758. doi:10.3390/
1047 ma15248758.
- 1048 [65] E. A. GIESS, J. P. FLETCHER, L. W. HERRON, Isothermal sintering of
1049 cordierite-type glass powders, *Journal of the American Ceramic Society*
1050 67 (8) (1984) 549–552. doi:10.1111/j.1151-2916.1984.tb19168.x.
- 1051 [66] N. M. PARIKH, Effect of atmosphere on surface tension of glass, *Journal*
1052 *of the American Ceramic Society* 41 (1) (1958) 18–22. doi:10.1111/j.
1053 1151-2916.1958.tb13497.x.
- 1054 [67] K. Watanabe, E. A. Giess, Crystallization kinetics of high-cordierite
1055 glass, *Journal of Non-Crystalline Solids* 169 (3) (1994) 306–310. doi:
1056 10.1016/0022-3093(94)90327-1.
- 1057 [68] Y. SUN, Z. YANG, D. CAI, Z. ZHANG, Q. LIU, S. FANG, L. FENG,
1058 L. SHI, Y. WANG, D. JIA, Crystallization kinetics, properties of α
1059 -cordierite based glass-ceramics prepared by glass powder sintering,
1060 *Journal of Inorganic Materials* 37 (12) (2022) 1351. doi:10.15541/
1061 jim20220179.
- 1062 [69] S. Wang, Y. Yu, Y. Zhao, P. Wang, J. Wang, Z. Yan, Q. Jing, Effect of
1063 na2o on the low-temperature densification, crystallization behavior, and
1064 dielectric properties of perlite tailings-derived α -cordierite glass-ceramics,
1065 *Materials* 19 (7) (2026) 1348. doi:10.3390/ma19071348.

- [70] Kyocera Corporation, Material properties: Ceramic packages / ceramic substrates (gl570, gl580, gl773 ltcc), Web page, <https://global.kyocera.com/prdct/semicon/material/>, accessed 24 September 2026 (2026).
- [71] J. U. Knickerbocker, F. L. Pompeo, A. F. Tai, D. L. Thomas, R. D. Weekly, M. G. Nealon, H. C. Hamel, A. Haridass, J. N. Humenik, R. A. Shelleman, S. N. Reddy, K. M. Prettyman, B. V. Fasano, S. K. Ray, T. E. Lombardi, K. C. Marston, P. A. Coico, P. J. Brofman, L. S. Goldmann, D. L. Edwards, J. A. Zitz, S. Iruvanti, S. L. Shinde, H. P. Longworth, An advanced multichip module (mcm) for high-performance unix servers, IBM Journal of Research and Development 46 (6) (2002) 779–804. doi:10.1147/rd.466.0779.
- [72] Aurubis, Technical datasheet c10200 (cu-of), 18 08 us, PDF, <https://www.aurubis.com/en/dam/jcr:90029b00-b968-486e-bde0-ce56f16022cd/c10200-cu-of-us.pdf>, accessed 24 September 2026 (2018).
- [73] National Physical Laboratory, Kaye and laby tables of physical and chemical constants, section 2.3.5 thermal expansion (version 1.1), Archived web page, https://web.archive.org/web/20190512035127/http://www.kayelaby.npl.co.uk/general_physics/2_3/2_3_5.html, accessed 24 September 2026 (2010).
- [74] G. R. Johnson, W. H. Cook, Fracture characteristics of three metals subjected to various strains, strain rates, temperatures and pressures,

- 1089 Engineering Fracture Mechanics 21 (1) (1985) 31–48. doi:10.1016/
1090 0013-7944(85)90052-9.
- 1091 [75] B. Banerjee, An evaluation of plastic flow stress models for the simu-
1092 lation of high-temperature and high-strain-rate deformation of metals,
1093 arXiv:cond-mat/0512466, arXiv:cond-mat/0512466 (2005).
- 1094 [76] P. L. Walker, F. Rusinko, L. G. Austin, Gas reactions of carbon, in:
1095 Advances in Catalysis, Vol. 11, Elsevier, 1959, pp. 133–221. doi:10.
1096 1016/S0360-0564(08)60418-6.
- 1097 [77] M. Barrio, B. Gøbel, H. Rimes, U. Henriksen, J. E. Hustad, L. H.
1098 Sørensen, Steam gasification of wood char and the effect of hydrogen
1099 inhibition on the chemical kinetics, in: Progress in Thermochemical
1100 Biomass Conversion, Blackwell Science, 2001, pp. 32–46. doi:10.1002/
1101 9780470694954.ch2.
- 1102 [78] P. L. Flaitz, A. M. Flanagan, J. M. Harvilchuck, et al., Method and means
1103 for co-sintering ceramic/metal MLC substrates, US Patent 5,130,067,
1104 issued 1992-07-14, assigned to IBM (1992).
- 1105 [79] D. J. Dubetsky, L. W. Herron, R. N. Master, Process for flattening glass-
1106 ceramic substrates, US Patent 4,340,436, issued 1982-07-20, assigned to
1107 IBM (1982).
- 1108 [80] S. Reinsch, M. L. F. Nascimento, R. Müller, E. D. Zanotto, Crystal
1109 growth kinetics in cordierite and diopside glasses in wide temperature
1110 ranges, Journal of Non-Crystalline Solids 354 (52-54) (2008) 5386–5394.
1111 doi:10.1016/j.jnoncrysol.2008.09.007.

- 1112 [81] DuPont Microcircuit Materials, Dupont greentape 951 low temperature
 1113 ceramic system, technical data sheet (mcm951, 7/2011), PDF, [https:](https://www.etsmtl.ca/uploads/Dupont_951.pdf)
 1114 [//www.etsmtl.ca/uploads/Dupont_951.pdf](https://www.etsmtl.ca/uploads/Dupont_951.pdf), accessed 24 September
 1115 2026 (2011).
- 1116 [82] Ferro Corporation, A6m/a6m-e high frequency ltcc tape system,
 1117 datasheet, PDF, https://www.etsmtl.ca/uploads/Ferro_A6M.pdf,
 1118 datasheet dated November 2015, accessed 24 September 2026 (2015).

1119 **Appendix A. Copper slab model and its calibration**

1120 The copper conductor uses the slab model developed for single-material
 1121 copper parts printed from a photocurable slurry. Besides the binder chemistry
 1122 of Section 2.2, it tracks the oxygen bound in the powder as Cu_2O . The oxide
 1123 grows in oxygen-bearing gas by a decelerating law fitted to powder oxidation
 1124 data, is reduced by hydrogen above the equilibrium ratio of Eq. (9), and
 1125 reacts with char by $\text{Cu}_2\text{O} + \text{C} \rightarrow 2\text{Cu} + \text{CO}$. None of these oxidation paths is
 1126 active in the co-firing programmes studied here, because the inlet gas contains
 1127 no oxygen and the water-to-hydrogen ratio is held below the copper–oxide
 1128 equilibrium. What matters for co-firing is the sintering viscosity. Eq. (7) uses
 1129 the lattice and grain-boundary diffusion data for copper tabulated by Frost
 1130 and Ashby [53], with one factor, f_η , fitted to the densification of copper parts
 1131 printed from a 60 vol% photocurable slurry of 22 μm powder and sintered
 1132 for 4 h at 1050 °C in hydrogen, which reached 90–94% relative density [32].
 1133 The fit gives $f_\eta = 0.28$, with 0.22–0.34 spanning the reported density range.
 1134 The powder size enters Eq. (7) through the grain size, set initially to 0.4
 1135 times the median particle size, and through the sintering stress. Transferring

1136 f_η from 22 μm to the 3–12 μm powders considered here therefore rests on
 1137 the diffusional scaling of Eq. (7); this is why f_η is included in the sensitivity
 1138 analysis of Section 4.10.

1139 **Appendix B. Verification**

1140 *Viscous densification of the glass-ceramic.* With crystallisation and char
 1141 removed, and open porosity, Eq. (3) with a Newtonian matrix gives $\dot{\theta} =$
 1142 $-9\gamma\theta/(4r\eta)$. The slab model was run through isothermal holds at 810, 830
 1143 and 850 $^\circ\text{C}$ with 4, 8 and 16 cells. The simulated porosity follows the closed
 1144 form, integrated along the simulated part temperature, to within 1.3×10^{-4}
 1145 over the full range of densification before pore closure, independently of the
 1146 mesh (Fig. 2a).

1147 *Bilayer curvature.* The three-dimensional model of a free 12×12 mm plate
 1148 made of 0.6 mm copper on 1.8 mm glass-ceramic was solved for the instan-
 1149 taneous velocity field at 900 $^\circ\text{C}$, and the curvature rate was fitted to the
 1150 mid-plane deflection. It converges to the viscous Timoshenko value of Eq. (10)
 1151 at least at second order: the error falls from 3.4% at 0.6 mm voxels to 0.69,
 1152 0.22 and 0.05% at 0.3, 0.2 and 0.15 mm (Fig. 2b).

1153 *Free sintering and mass conservation.* A free, frictionless cube held isother-
 1154 mally densifies homogeneously and isotropically, matches the closed-form
 1155 Skorohod–Olevsky solution to 0.22% in density, and conserves solid mass to
 1156 0.07%.

1157 *Crystallisation kinetics.* With the rate constant calibrated as described in
 1158 Section 2.3, the simulated exotherm at 10 K min^{-1} peaks within 0.01 K of

¹¹⁵⁹ the prescribed temperature for Avrami exponents of 1, 2 and 3.